

OneERP Program
Development Guideline
for classic S/4 HANA systems



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22.0	02.-08.2025	Uwe Nüsse	Add new rules for Preussen Elektra Rule definition for Number Range Objects Accept the SAP module CCM (SAP Contact Center Management) as own module. Accept the SAP module MDG (Master Data Governance) as own module. New rule for the usage of SAP GUI Controls Development approach assessment Changes in forbidden technique area Contribution to support the SAP clean core approach

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Quality Review

The following have been nominated as quality reviewers for this document:

Name	Title	Date of Issue	Versions

Approval/Sign-off

This document requires approval from the following individuals:

Name	Role	Date	Signature

Content

1. Introduction	9
2. Realization	10
2.1. ONEERP SYSTEM LANDSCAPE	10
2.2. SAP NAMESPACES FOR NEW E.ON	11
2.3. DEVELOPMENT APPROACH ASSESSMENT	12
2.4. DEVELOPMENT RULES	14
2.4.1. DEVELOPMENT PACKAGES AND TRANSPORT LAYER FOR ABAP	14
2.4.2. DEVELOPMENT SEPARATION (SEPARATION OF CONCERNS)	16
2.4.3. CLONE PROGRAMS AND OTHER SUB OBJECTS	17
2.4.4. PATTERN	17
2.4.5. IMPORT ERROR CONTROL AND DOWNGRADE CHECK	18
2.4.6. PROXY DEVELOPMENTS	19
2.4.7. OTHER GENERAL RULES	21
2.4.8. DESIGN	23
2.4.9. SOURCE CODE MANAGEMENT	23
2.5. THIRD PARTY APPLICATIONS	23
2.6. QA TOOL USAGE AND INTEGRATION INTO THE TRANSPORT RELEASE	25
2.6.1. CONTROL FOR CODE (C4C)	25
2.6.2. INTEGRATION INTO THE TRANSPORT RELEASE IN SOLUTION MANAGER (OCC)	26
2.6.3. EXEMPTION PROCESS	26
2.9. NAMING CONVENTIONS	27
2.9.1. SOURCE CODE FORMAT CONVENTIONS	27
2.9.2. NAMING CONVENTIONS FOR REPOSITORY OBJECTS	29
2.9.3. TRANSLATION RELATED RULES	33
2.10. FIORI APP PROGRAMMING	33
2.11. DEVELOPMENT PACKAGES AND TRANSPORT LAYER FOR FIORI	35
2.12. S4U - NAMING CONVENTIONS FOR EMBEDDED BI/BW	36
2.13. NAMING CONVENTIONS FOR DATA DECLARATIONS ENHANCEMENTS	38
2.14. MODIFICATIONS AND OTHER FORBIDDEN TECHNIQUES	40
2.15. TABLE AND STRUCTURE ENHANCEMENTS	41
2.15.1. CONVENTIONAL OR STANDARD ENHANCEMENT WITH CI_INCLUDES	41
2.15.2. ENHANCEMENT WITH CUSTOM FIELDS & LOGIC	44
2.15.3. INDEXES UNDER SAP HANA	44
2.16. GDPR ENHANCEMENT	45
2.16.1. FUNCTIONAL REQUIREMENT	45
2.16.2. TECHNICAL IMPLEMENTATION	45
2.16.2.1. VENDOR DATA PROTECTION	47
2.16.2.2. CUSTOMER DATA PROTECTION	49

2.16.2.3.	BUSINESS PARTNER DATA PROTECTION.....	50
2.16.2.4.	PROTECTION OF DATA USED IN MASS-PROCESSES.....	51
2.17.	OBSOLETE PROGRAM STATEMENTS	53
2.18.	UNLIMITED USAGE OF PROGRAMMING COMMANDS.....	53
2.19.	EXITS	54
2.19.1.	PROGRAM EXIT - FRAMEWORK.....	54
2.19.2.	PROGRAM EXIT – IN TOP-INCLUDES.....	57
2.19.3.	PROGRAM EXIT – IN OTHER INCLUDES	58
2.19.3.1.	USER-EXIT IS A MU-SPECIFIC OPERATIONS IMPLEMENTATION	58
2.19.3.1.1.	NAMESPACE FOR OBJECTS CREATED IN THE DEO SYSTEM	58
2.19.3.1.2.	NAMESPACE OF OBJECTS IN THE ZX* INCLUDE	58
2.19.3.1.3.	CODING IN OPERATIONS-INCLUDE	58
2.19.3.1.4.	STEERING TABLE	58
2.19.3.2.	USER-EXIT IS CENTRALLY IMPLEMENTED IN THE COMMERCIAL CORE SYSTEM.....	60
2.19.3.2.1.	INTERFACE DEFINITION.	61
2.19.3.2.2.	MAINTENANCE VIEW AUTHORITY CHECK – TEMPLATE PROTECTION	63
2.20.	SCREEN EXIT.....	65
2.20.1.	PART SCREEN PROCESS LOGIC.....	66
2.20.2.	PROGRAM EXIT LOGIC.....	67
2.21.	MENU EXIT	67
2.22.	FIELD EXIT.....	67
2.23.	CUSTOMER SCREEN CONTROL FRAMEWORK	67
2.24.	CMOD-DOCUMENTATION	68
2.25.	ENHANCEMENTS.....	68
2.26.	AUTHORIZATION/PERMISSION	70
2.26.1.	AUTHORIZATION/PERMISSION GROUP	70
2.26.2.	AUTHORIZATION/PERMISSION OBJECT	70
2.26.3.	PREPARED DEVELOPER ROLES	70
2.27.	OTHER NEW DEVELOPMENT TOOLS AND AREAS	71
2.27.1.	EEWB (EASY ENHANCEMENT WORKBENCH).....	71
2.27.2.	AET (APPLICATION ENHANCEMENT TOOL)	71
2.27.3.	AIF (APPLICATION INTERFACE FRAMEWORK).....	71
2.27.4.	BRF+ (BUSINESS RULE FRAMEWORK PLUS).....	71
2.27.5.	CDS VIEWS (CORE DATA SERVICES)	72
2.27.6.	ABAP IN SAP CLOUD PLATFORM (SCP OR BTP).....	76
2.27.7.	JAVASCRIPT RECOMMENDATIONS	76
2.27.8.	JAVA RECOMMENDATIONS.....	81
2.28.	INTERNET TRANSACTION SERVICES (ITS)	83
2.28.1.	PREREQUISITES	83
2.28.2.	CREATION OF THE INTERNET TRANSACTION SERVICE WITH SE80	83
2.28.3.	DEFINITIONS IN THE TRANSACTION SICF	86

2.28.4.	CHANGES ON SCANNER SIDE.....	89
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1. Introduction

The purpose of this document is to articulate a common and mutually valid set of standards and procedures for naming, development and documentation of ERP SAP ABAP – applications with the goal of maximizing the quality, value, security and maintainability of new or existing ABAP development. It is intended as a resource for all ABAP development team members from developers to quality managers delivering solutions for the Group both internally and externally.

Standardized ABAP coding is an essential factor for success in an efficient processing of support requests in the various systems for various application areas and customers. Furthermore, there are legal and contractual requirements to be obeyed.

In addition, security has high criticality in SAP environments. Connections into the web and the opening for Digital Transformation give space for scenarios to penetrate a system from outside of the Corporate Network.

The usage of an automated and standardized Code quality check is necessary to fulfill the different rules and recommendations from standard organizations.

In other words, by adhering to the rules stated in this document, better code security, readability, maintainability and reusability across the Group are intended to be achieved.

This guideline is based on the following guidelines from the participating source companies:

INNOGY:

- S4i ABAP-Development-Standards-2018-02-08

E.ON:

- SAP_Development_Guidelines_7.0_published_06102017
- Additional_GLOBE_Development_Guideline_V 2.04

Project:

- Updated Guideline from BI Team

This guideline replaces the previous version and becomes valid on the day of its publication in Teams.

2. Realization

2.1. oneERP system landscape

The future S/4 system landscape contains ERP systems for different purposes and is currently defined as follows. Because this landscape is subject to constant change due to the requirements of migration, the following graphic has been removed and reference has been made instead to the current version in Change Management Portal.

The landscape attached to the portal is an overview of the most important systems and stages. The latest version of the distribution landscape is always available here [S4U-/EAM-portal](#).



HEADLINE NEWS:	
Current system issues	None
Current on-going system work	None
Weekly non-prod maintenance:	Nothing currently planned
Monthly production maintenance:	Nothing currently planned
TDO data refresh:	QEC/QEO data will be refreshed with data from MC3/MO3 (excluding customizing tables). This requires outage (all users will be locked and all UC4 jobs stopped) of QEC/QEO From 14:10@16:00 CET to 17:10@13:00 CET
Planned System/Client integration or refreshes:	Two system refreshes are planned for 16.09.22 - 21.09.22: MC0 copied from QEC MO0 copied from QEO MC0, MO0 Integration into CharM 23.09 17:00 - 26.09 12:00 CharM outage during this time. PEO->MO4 and PEC->MC4 refreshes planned for October
Fast Close - financial closing first Swds after quarter end	No regular Production Release. Business Customizing, Fast Track or Business Critical changes only to Production Q3 03/10/2022 to 07/10/2022, Q4 03/01/2023 to 09/01/2023

In addition, there is one Development Transfer System (DET) defined as a standalone system. MDG and SSF functionalities are to be developed in the separate development system DEA/100 (parallel to DEC/100).

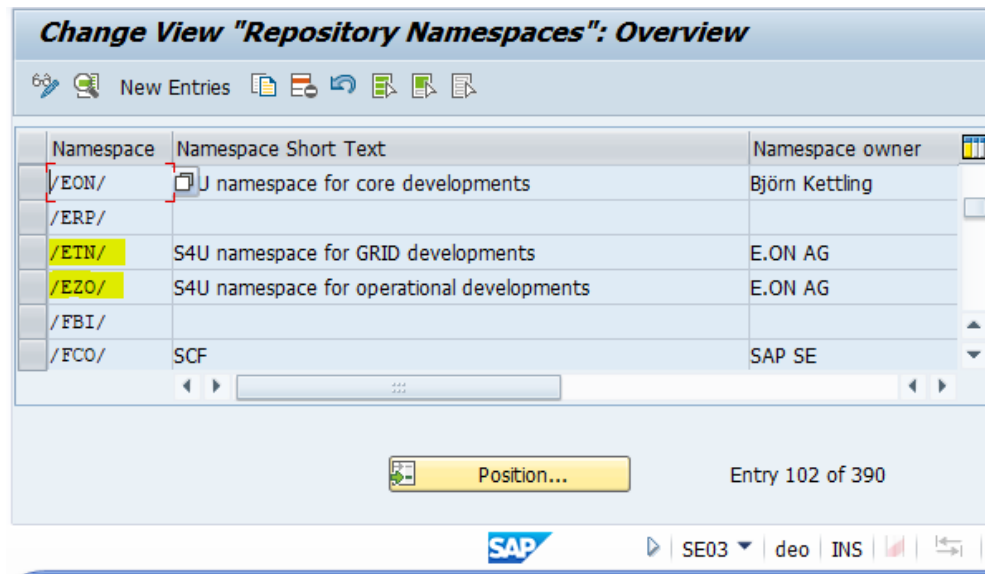
This guideline does not apply for developments in IS-U or CRM systems.

2.2. SAP namespaces for new E.ON

SAP offers the possibility to license customer own namespaces. The following namespaces are licensed for the new system landscape and purposes. When developing workbench objects one of the following suitable development namespaces must be chosen as per this table:

Development System	Namespace	Description
DEC	/EON/ & /ETS/	/EON/ covers all core developments in the core development systems DEC, D00, DGR. /ETS/ covers all developments of the Innogy ETS-ADÜ Tool for oneERP for asset accounting
DEA	/EZW/	covers all core add on developments in the MDG development system DEA
DEO	/EZO/	covers all operations related developments in the development system DEO
DEO, YEO	/ETN/ & /KAW/ & /INCC/	covers all network/grid related developments in the development system DEO plus /INCC/ for the special application Internal-& Cross-Company Clearing

Respective SE03 namespace settings in DEO/100:



Change View "Repository Namespaces": Overview

Namespace	Namespace Short Text	Namespace owner
/EON/	U namespace for core developments	Björn Ketting
/ERP/		
/ETN/	S4U namespace for GRID developments	E.ON AG
/EZO/	S4U namespace for operational developments	E.ON AG
/FBI/		
/FCO/	SCF	SAP SE

Position... Entry 102 of 390

SAP SE03 deo INS

Namespace	Namespace Short Text	Namespace
/ILE/	Label Localization for ERP (Annexing)	SAP AG
/ILM/		SAP SE
/INCC/	EAM IM/CC	E.ON AG
/INCMD/	Interchangeability Master Data	SAP SE
/IPRO/	Document Builder	SAP AG
/ISDFPS/	EA-DFPS	SAP AG
/ISHCDP/		
/ISHFR/		
/ISIDEX/	IDEX	SAP AG
/IWBEP/		
/IWCOR/		
/IWFND/		
/IWHDB/		
/IWNGW/		
/IWPGW/		
/IWRK/		
/IWXBE/	Exchange of Events	D037843
/IXOS/	OpenText Development ADA	OPENTE
/J7L/	REA Recycling Administration German	SAP SI
/KAW/	LEW KA-Wesen	E.ON AG
/KAYAK/		

Position... Entry 133 of 391

SAP SE03 deo INS

Important: Development objects in customer own namespaces **must not** have the original system "SAP", to avoid impacts to the SPDD / SPAU during patching.

The /INCC/ namespace is a specific namespace for all ABAP Developments for the automated creation of bid, accounting and billing documents depending on the contractual arrangements within the companies of a REVU/REG Holding. Objects from that namespace must not be used in developments from other namespaces and objects from other namespaces must not use /INCC/ objects. This is necessary because the projects can have different distribution cycles and project plans.

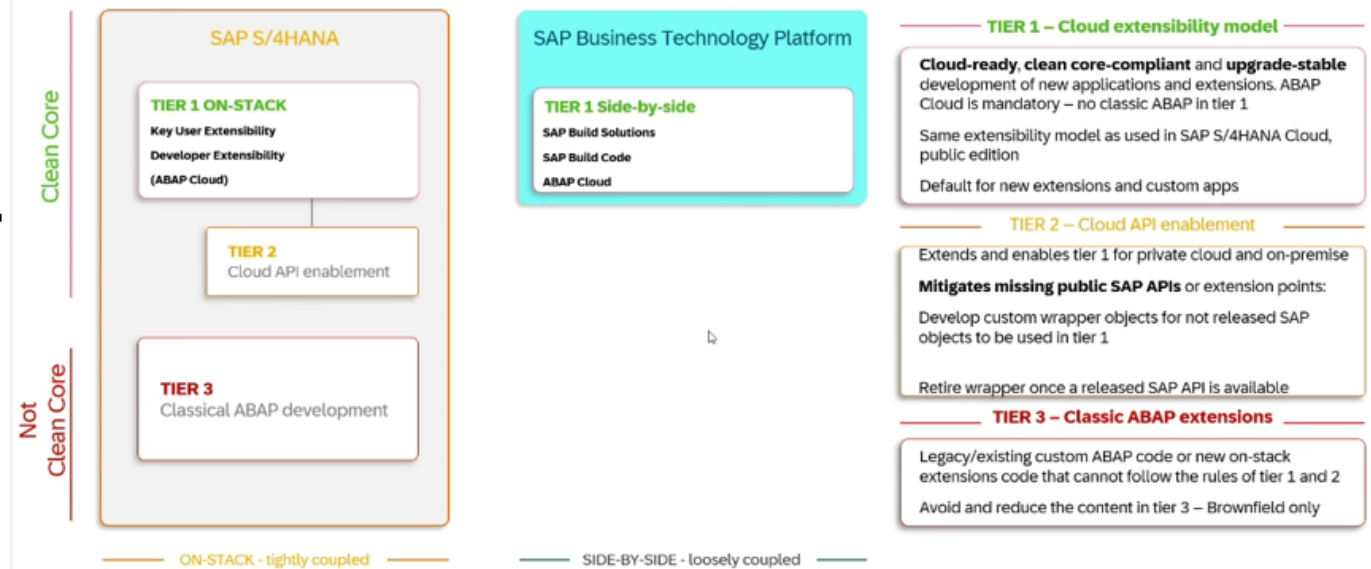
Only exception is the insert of control table records for Non-Scope implementations in Scope exit control tables (R3TR TABU /EON/* in DEA/DEO transports). See the details in the respective chapter further below.

The /AKB/ namespace is for developments of the BKZ application from former Innogy systems. This application is defined as part of the corporate template by the financial accounting product owner. Till 2026 its original remains in the ESM environment. After 2026 it will become part of the corporate template.

2.3. Development approach assessment

New implementations are strongly recommended to be implemented in a public cloud compatible way.

3-tier extensibility model for SAP S/4HANA private cloud and on-premise



2.4. Development Rules

2.4.1. Development Packages and Transport Layer for ABAP

For the distribution of developments is the assignment to development packages essential.

Packages are generally named as follows: Name = /NS/XXPP

NS	Namespace or Y or Z
XX	Application component or module
PP	Free definition

As soon as SAP restricts us in using our defined namespaces for package names, also Y or Z names are allowed.

The same rules apply for generated objects and interface parameter, if the interface from/to a third-party application would not work as designed anymore. Example:

/EON/LCAVIM_IDOC\$04	8	Naming Conventions	Invalid name PIS_IDOC_CONTROL for IMPORTING parameter ...
/EON/LCAVIM_IDOC\$05	8	Naming Conventions	Invalid name PIS_IDOC_CONTROL for IMPORTING parameter ...
/EON/LCAVIM_IDOC\$09	8	Naming Conventions	Invalid name INPUT for IMPORTING parameter (FUNCTION)

The application component or module names are based on the structure that **transaction /n/HIER** provides from an SAP system.

Exceptions:

- Document Management (SAP module CA-DMS): We accept the naming convention for such objects to begin with /EON/CADMS.
- CATS: The responsible team decided to name their objects /EON/CATS.
- GRC: the GRC team is allowed to use the modules "AC" for Access Control and "PC" for Process Control developments and dictionary objects.
- the central teams working on that are allowed to use the module "LR" for developments and dictionary objects from the application LoKaRef (Location References or Lokationsreferenz).
- For the SAP Module CCM (SAP Condition Contract Management) is the naming convention /EON/CCM accepted.
- We accept also as naming convention for modules "MDG" for Master Data Governance objects.

- Warehouse Management: for the warehouse management we accept the module name value "WM". In HIER it is called:

HLA0009518 LE-WM Warehouse Management

Y is to be used for Template Core packages developed in DEC (Example: SMTE in DEC, for which we use the package YSDBS_MDG)

Edit Mapping Step: SALES_ARRANGEMENT (Extension Mode)

Mapping BPBUPA_RPLRQ_ECC_CU_IN > Mapping Step SALES_ARRANGEMENT
Mapping Description BPBupa Repl Req Customer Inbound Mapping in ECC Package Group ERP_MDG_SI

Please select a transformation
Object SMTE 000D3AAC85A01EEAA9C7641BFAD3B1BB cannot be assigned to package /EON/SDBS_MDG - [Display Help](#)

Save Close Read-Only Copy Delete Mapping External

Extension
* Owner: Customer Enhancement
Switch:

Field Checks Transformations

Order	*Transformation Type	Conditional
00001	Field Mapping	☐
00002	Complex Transformation	☐
00003	Complex Transformation	☐
00004	Complex Transformation	☐
00005	Complex Transformation	☐

Tree structures are available to the user to automatically Assign or Map structure fields to a Field Check, a

Source Structure

Help

Object SMTE 000D3AAC85A01EEAA9C7641BFAD3B1BB cannot be assigned to package /EON/SDBS_MDG

Message no. TO143

Diagnosis

Object SMTE 000D3AAC85A01EEAA9C7641BFAD3B1BB is not in namespace /EON/. It therefore cannot be assigned to package /EON/SDBS_MDG.

System Response

Editing is terminated.

Procedure

Assign the object to a package whose name does not begin with a namespace prefix. The first character of the name of the package must not be "/".

OK

Z package names are to be used in the non-core areas and in SMOD/CMOD as predefined by SAP. Further details are given further below in that document. This applies as well for objects, which are called from central E.ON systems outside of S/4 United, like central HR system P83 or central printer control system P5A.

The following **central S/4 United transport layer** are defined.

DEC (core)	=>	ZCOR	
DEC (BKZ)	=>	ZET5	for Third Party Namespace /AKB/
DEC (IRMA)	=>	ZTIV	for Third Party Namespace /IKOR/
DEA (core – add on)	=>	ZADD	

DEO, YEO (operations)	=>	ZOPS	for Namespace /EZO/
DEO, YEO (network/grid)	=>	ZNET	for Namespaces /ETN/ and /KAW/ and /INCC/
DET (transfer system)	=>	ZUWE	

For complex developments, the standard mentioned above may be extended in order to create several corresponding packages. In this case, please use an underscore to separate the extension from the first part of the package name.

Example:

- /EON/MMWF
 - /EON/MMWF_FORMS
 - /EON/MMWF_APPEND

Please consider using a package hierarchy in these cases.

The pre-definition in the degree of detail is limited to the top 3 layers. That will avoid the fragmentation which will not be possible to manage anymore.

During transfers from old systems to the new S4U environment are not only namespace changes the duty of the developer, but also the respective changes of the original systems of a development object. This has to be done before editing an object in the new system, to avoid the creation of transport tasks with type "Repair". Repairs pop-up again in upgrade processes (SPDD, SPAU) and cause issues there.

Inactive objects are to be checked and activated in every system. Problems have to be solved by the implementing developer.

2.4.2. Development separation (Separation of Concerns)

All newly created objects for one functionality have to be assigned to the same development package or sub-package during the development phase.

The assignment of an object to one development package follows the strategy to create a structure which encapsulates the different developments, and to make them transportable as isolated groups. Therefore also redundancies of defined objects are accepted. A disregard of this convention will lead on one hand to bigger functionality mixtures. On the other hand, it can lead to system inconsistencies (like transportation import errors or short dumps) and/or an increased effort, time and number of iterations for bug fixing and transportation tasks.

Objects from foreign developments are forbidden to be used, because in this case the developer will not be independent in his use anymore. He will depend on other developers in that case, that he has to align his requirements, progress and transport releases with others. Furthermore, he has a higher risk for his own functionality, in cases of later changes of the used foreign object. The adherence of own objects to foreign development packages raise the risks for other developers and functionalities in the same way and is therefore forbidden.

The same regulations are valid for other structural levels, like function groups, programs, modification projects or classes.

2.4.3.Clone programs and other sub objects

To copy or clone a SAP standard functionality is forbidden for the future. Each clone encapsulate our functionality from the progress SAP provides with support packages and notes. Incomplete clones cover also the risks of internal interface incompatibilities leading to dumps after upgrades. Before creating a clone has the functional designer, the IT coordinator and the developer the duty to check whether the functionality changes cannot be integrated into the SAP standard with the help of the various customer function, Exit and other enhancement functionalities. The need to nevertheless clone a SAP standard program has to be confirmed also by the respective Solution Architect(s), similar to the business function and system enhancement process.

2.4.4. Pattern

All customer own development objects have to have at least a header with comments about the following aspects:

```
*****
* Brief description of the function:
* (Update of the service package characteristic for equipment in this
*   class Sockets dependent on the Property characteristic)
* (Read and/or write access to the data?)
* (Access procedure for modifying access?)
* (Technical dependencies?)
* (Performance requirements?)
* (Sequence control?)
* (Copy of #?)
*
* SAP Release: = XXX X.X
*****|
```

They furthermore have to have a change history with information about the date, the user and the reason for the change:

```
*****
* Change history:
* Date      User      Reason(Description and CR/ORDER/TICKET ID)
* -----
* DD.MM.YY XXX      XXX
*
*****
```

A CONSTANTS definition for the global constant gc_tcode has to be defined either in the Top-Include or after the change history of the object.

CONSTANTS: gc_tcode TYPE tcode VALUE '/NS/XX_PP'.

After Initialization or Start-of-selection has an authority check to be defined in each objects.

```
INITIALIZATION or Start-of-Selection.  
  AUTHORITY-CHECK OBJECT 'S_TCODE'  
  ID 'TCD' FIELD gc_tcode.  
  
  IF sy-subrc <> 0.  
    MESSAGE a172(00) WITH gc_tcode.  
  ENDIF.
```

This can be programmed directly in the report source code or can be excluded by calling a predefined generic function. Programs in customer namespaces provided by SAP in Notes have to have the above mentioned pattern and a transaction for the execution in front.

During the implementation of new or the correction of existing executable objects is the alignment with the IAM team to be initiated by the developer or development team lead. This has to happen

- before the initial functional test in the CEC/CEO system takes place.
- Based on a modifiable workbench transport.
- For project changes (S4U, EAM, ...) via email to the mailbox S/4 UNITED-IAM s4_united-iam@eon.com.
- For changes from the maintenance lines via ticket in Service Now (SIAM) <https://siam.service-now.com/esp/>
→ raise an incident. The assignment group of the incident must be: eon-all-sap-authorization-service.

This is important to maintain the SU24 according to the requirement of the businesses.

2.4.5.Import Error Control and Downgrade Check

The distribution of a change into a target system (DE*, CE*, QE*, PE*), based on Solution Manager triggered transports or manual cross transports, takes place only one time and defines the following responsibilities for the transport owner and the assigned developer:

- Check the transport return code per system/client.
- In case of RC greater/equal 8
 - o Evaluate the error consequences for the system (e.g. follow up customizing cannot be imported completely, transactions/program are inconsistent and will dump during execution, ...)
 - o Define a proper solution for the correction

- Implement and apply the defined solution with a new follow up transport.

It is also the responsibility of the developer to check the transport content of a Work Item or Defect Correction for any downgrade risks listed in the Solution Manager WEB UI. It appears to be displayed here:

STATUS OVERVIEW

RELATED TRANSACTIONS

1

2

3

4

5

6

7

8

9

Created

In Development

To Be Corrected

To Be Tested

Successfully Tested

Handed Over to Release

Preliminary Import Requested

Test for Preliminary Import

Tested for Production Import

DETAILS

TEXT

RICH TEXT

ATTACHMENTS

TRANSPORT MANAGEMENT

TRANSPORT-RELATED CHECKS

Perform check

Refresh

Status	Type	Result
	Downgrade Protection	Very critical conflicts: 0; critical conflicts: 0; warning conflicts: 0 (including history)

2.4.6. PROXY Developments

The creation of a PROXY service requires some objects from the customer namespace and system. During the creation appears the following Pop-Up to be displayed.

Enter Package/Request

Configure Service

Complete

Enter the package, prefix and transport request for the proxy object.

Enter Package & Prefix

Package

Prefix

Request/Task

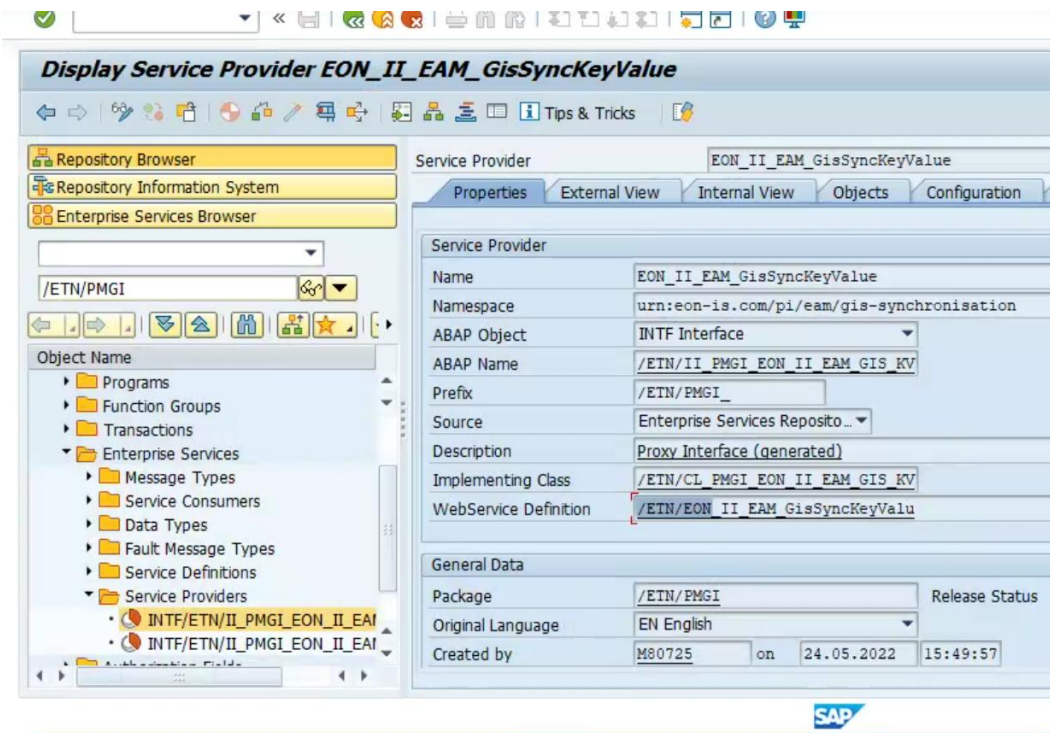
Local Object

The entries in this Pop-Up must take these rules into account:

- The development package name has to respect the already defined rules for packages within this guideline.
- The Prefix must have the following name structure: /NS/xyyy... (xx = module, yy... = free text)
- The transport request/task must be assigned to a Work Item, Defect Correction or Change Document created in the Solution Manager System PSM/100.

After maintenance of this Pop-Up and pressing continue follows the detail creation/generation screen.

Example screen from DEO/100:



The Interface ABAP-name must begin with the E.ON Namespace. The valid namespaces for E.ON in the different S4U/EAM development systems are defined already above in the guideline. The naming convention is defined as: /NS/II_xyyy... (xx = module; (yy...= free text))

The implementing class name has to follow the already defined rules for classes within this guideline.

The WEB service definition must begin with /NS/EON_II_yyy... (yy...= free text)

Type	ABAP Name	Prefix	Name	Namespace
	/ETN/II_PMG1_EON_II_EAM_FMBASE	/ETN/PMGI_	EON_II_EAM_FMBASE_KeyValue	urn:eon-is.com/pi/eam/FMBASE
	/ETN/PMGI_EON_MT_EAM_FMBASE_KE	/ETN/PMGI_	EON_MT_EAM_FMBASE_KeyVal...	urn:eon-is.com/pi/eam/FMBASE
	/ETN/CX_PMG1_EON_MT_EAM_FAULT1	/ETN/PMGI_	EON_MT_EAM_FaultMessage	urn:eon-is.com/pi/eam/Comm...
	/ETN/PMGI_EON_DT_EAM_FMBASE_K7	/ETN/PMGI_	ENTRIES	
	/ETN/PMGI_EON_DT_EAM_FMBASE_K6	/ETN/PMGI_	ENTRY	
	/ETN/PMGI_EON_DT_EAM_FMBASE_K8	/ETN/PMGI_	EON_DT_EAM_FMBASE_KeyVal...	urn:eon-is.com/pi/eam/FMBASE
	/ETN/PMGI_EXCHANGE_FAULT_DATA1	/ETN/PMGI_	ExchangeFaultData	urn:eon-is.com/pi/eam/Comm...
	/ETN/PMGI_EXCHANGE_LOG_DATA	/ETN/PMGI_	ExchangeLogData	urn:eon-is.com/pi/eam/Comm...
	/ETN/PMGI_EON_DT_EAM_FMBASE_K5	/ETN/PMGI_	HEADER	
	/ETN/PMGI_EON_DT_EAM_FMBASE_K4	/ETN/PMGI_	ITEM	
	/ETN/PMGI_EON_DT_EAM_FMBASE_K3	/ETN/PMGI_	ITEMS	
	/ETN/PMGI_EON_DT_EAM_FMBASE_K2	/ETN/PMGI_	MESSAGE_HEADER	
	/ETN/PMGI_EON_DT_EAM_FMBASE_K1	/ETN/PMGI_	RETURN	
	/ETN/PMGI_EON_DT_EAM_FMBASE_KE	/ETN/PMGI_	RETURNS	
	/ETN/PMGI_EON_DT_EAM_FMBASE_TAB2	/ETN/PMGI_	ENTRY	
	/ETN/PMGI_EXCHANGE_LOG_DAT_TAB	/ETN/PMGI_	ExchangeLogData	urn:eon-is.com/pi/eam/Comm...
	/ETN/PMGI_EON_DT_EAM_FMBASE_TAB1	/ETN/PMGI_	ITEM	
	/ETN/PMGI_EON_DT_EAM_FMBASE_TAB	/ETN/PMGI_	RETURN	

Generated table and structure names by the system can remain unchanged.

Typ	ABAP-Name	Präfix	Name	Namensraum	Status	Paket
	/ETN/II_PMG1_EON_II_EAM_GIS_KV	/ETN/PMGI_	EON_II_EAM_GisSyncKeyValue	urn:eon-is.com/pi/eam/gis-synchronisation		/ETN/PMGI
	/ETN/CX_PMS_GIS_EON_MT_EAM_FAU	/ETN/PMS_GIS_	EON_MT_EAM_FaultMessage	urn:eon-is.com/pi/eam/gis-synchronisation		/ETN/PM_GIS
	/ETN/PMGI_EON_MT_EAM_GIS_INPU	/ETN/PMGI_	EON_MT_EAM_GisSyncKeyValue	urn:eon-is.com/pi/eam/gis-synchronisation		/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_ENTRS	/ETN/PMGI_	ENTRIES			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_ENTR	/ETN/PMGI_	ENTRY			/ETN/PMGI
	/ETN/PMGI_EON_MT_EAM_GIS_SYN	/ETN/PMGI_	EON_DT_EAM_GisSyncKeyValue	urn:eon-is.com/pi/eam/gis-synchronisation		/ETN/PMGI
	/ETN/PMS_GIS_EXCHANGE_FAULT_DA	/ETN/PMS_GIS_	ExchangeFaultData	urn:eon-is.com/pi/eam/gis-synchronisation		/ETN/PM_GIS
	/ETN/PMS_GIS_EXCHANGE_LOG_DATA	/ETN/PMS_GIS_	ExchangeLogData	urn:eon-is.com/pi/eam/gis-synchronisation		/ETN/PM_GIS
	/ETN/PMGI_EON_DT_EAM_GIS_HD	/ETN/PMGI_	HEADER			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_ITM	/ETN/PMGI_	ITEM			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_ITS	/ETN/PMGI_	ITEMS			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_MHD	/ETN/PMGI_	MESSAGE_HEADER			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_RET	/ETN/PMGI_	RETURN			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_RETs	/ETN/PMGI_	RETURNS			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_TENTR	/ETN/PMGI_	ENTRY			/ETN/PMGI
	/ETN/PMS_GIS_EXCHANGE_LOG__TAB	/ETN/PMS_GIS_	ExchangeLogData	urn:eon-is.com/pi/eam/gis-synchronisation		/ETN/PM_GIS
	/ETN/PMGI_EON_DT_EAM_GIS_TITM	/ETN/PMGI_	ITEM			/ETN/PMGI
	/ETN/PMGI_EON_DT_EAM_GIS_TRET	/ETN/PMGI_	RETURN			/ETN/PMGI

The re-use of existing Fault Message, Exchange Fault Data Exchange Log Data structures and tables is not allowed, to avoid dependencies to and overtaker of other implementations.

2.4.7. Other general Rules

Some further and more general statements as far as programming rules are concerned. [This document does not differ between Business Units to keep it as generic as possible.](#)

Topic	Advice
Authority check	For every program an authority check has to be defined. Authorization checks must be implemented in order to protect the data. This means that, for example, in the case of selections of company code data, the authorizations for reading the company codes must be checked before access. This applies also for all other org.elements.
Pretty Printer	Use pretty printer.
Language	Development Language is always English. Register language-dependent data in the language-dependent tables. Consider the language related rules in the chapter "Translation related rules".

Constants	Use constants instead of literals.
Fixed Texts in messages	Fixed texts are forbidden. Constants to be used instead.
Macros	Macros are not allowed.
OK_CODE	An OK_CODE variable must be specified in all screens. This variable is defined via the field list of each screen created (exists as the last field in the field list).
Export parameters	All export parameters are to be initialized before their first use (CLEAR, REFRESH or FREE) in an encapsulated routine.
Database load	Reduce it as much as possible (avoid reading tables repeatedly, sort data in program, ...).
Database Selects	The selection of the required data from the database must be reduced to a minimum. Ideally it is done only once and then temporarily stored in an internal global sorted table. After that, can the value be determined much faster from there (via the key). To be checked in the CEC or CEO, based on the TOC with Transaction ST12 and the ABAP & SQL Trace at the latest.
Generated source code	Must not be changed.
Customer objects based on SAP notes or SAP Incidents	They have to follow this guideline and must contain at least: • an authority check if it is a report. • a Dequeue / Enqueue logic before DB changes • a description and change history (include the OSS Note or Incident number).
Modifications	Follow the defined modification process and must be approved by the Solution Architect and the System Owner. The approved modification request must be attached to the solution manager work item.
Generated objects	The name SAP defines during generation cannot be changed and needs to be accepted. Example: CDS view /EON/AA9F2B9B93C from DEC, which belongs originally to MM.
Limits of Functional Method Calls	As soon as more than one return parameter value (e.g. one return parameter value plus one classical exception value) is possible in a called method, a functional method call is not allowed. The reason for that is, that the classical exception handling is mandatory, but not possible after a functional method call.
n.n	n.n

As soon as applicable for the own ABAP development, are the principles, described in the "CSD-20 Secure Application Development (Draft)" document to be considered in the design, development and testing phases.

Link: [CSD-20 Secure Application Development \(Draft\) - Cyber Security - Confluence \(eon.com\)](#)

2.4.8. Design

Security precautions are part of the design phase in the software development to identify threats, plan for countermeasures and document the application's trust boundaries, components and significant data flows.

The Security by Design (SbD) principles **MUST** continuously be considered in all phases and iterations of the software development lifecycle.

Threat modelling **SHOULD** continuously be performed in all phases and iterations of the software development lifecycle.

2.4.9. Source Code Management

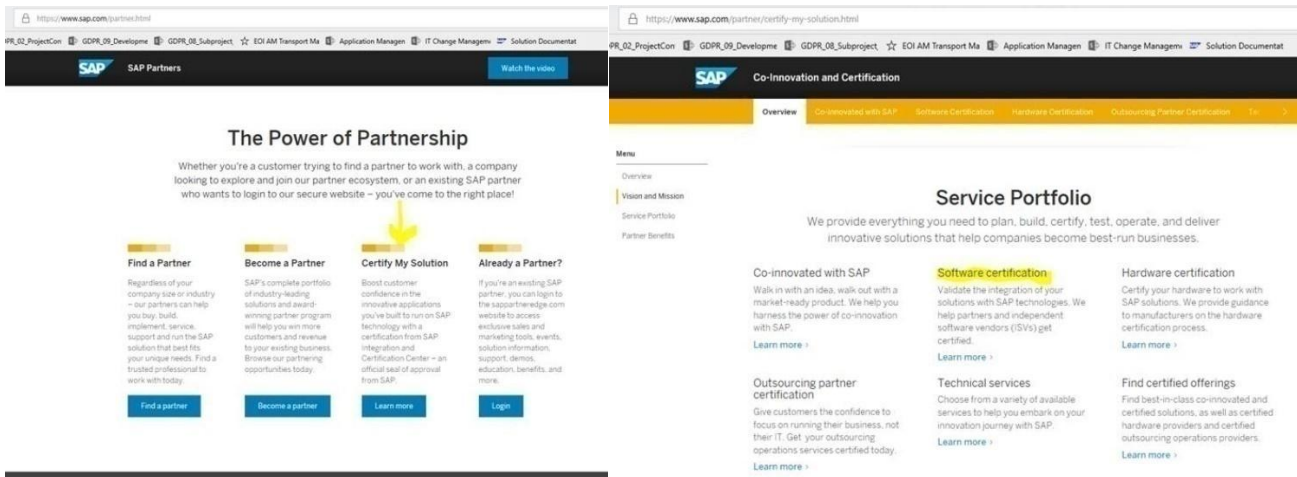
The foundation of every software development is the source code, that is tracked and versioned by using a Source Control Management (SCM). It must be ensured that changes to the source code take place in a secure and controlled manner.

- Central managed GitLab instance **SHOULD** be used as SCM.
- Source Code **MUST** exist in central GitLab before a deployment to a productive environment or system.
- The developer **SHOULD** commit only to non-productive branches.
- Secrets/API-keys/Access tokens **MUST NOT** be committed into the repository.
- Requests for merging of code changes or production deployment triggers **SHOULD** be approved by at least one person that is different from the requester (four-eyes-principle).

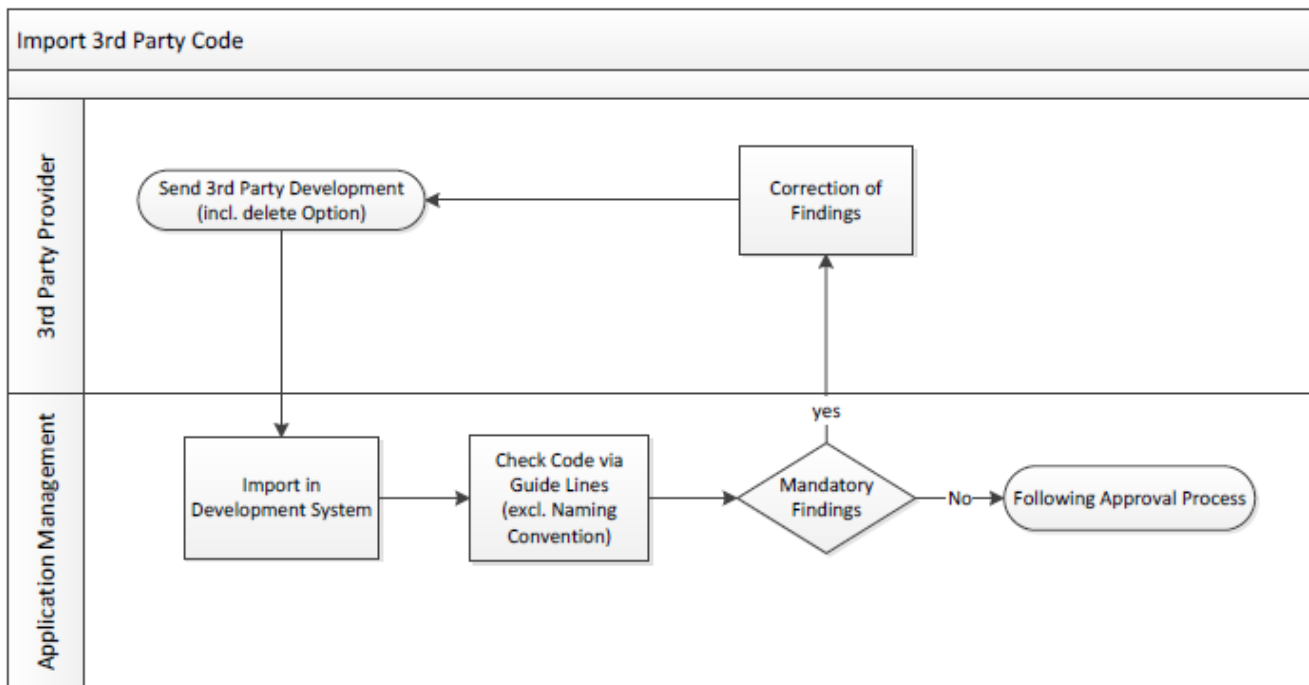
2.5. Third Party Applications

Software from third party application provider are to be applied thru the entire S/4 United landscape with the original transports to make clear, that they were not created in the own development systems. The provider is to be requested to provide a SAP certificate about the release compatibility and security against external attacks.

Where can I get a certificate from [SAP](#) for my third party application? See here:



This can be given also from a SAP certified provider, like Onapsis (ex. Virtual Forge).



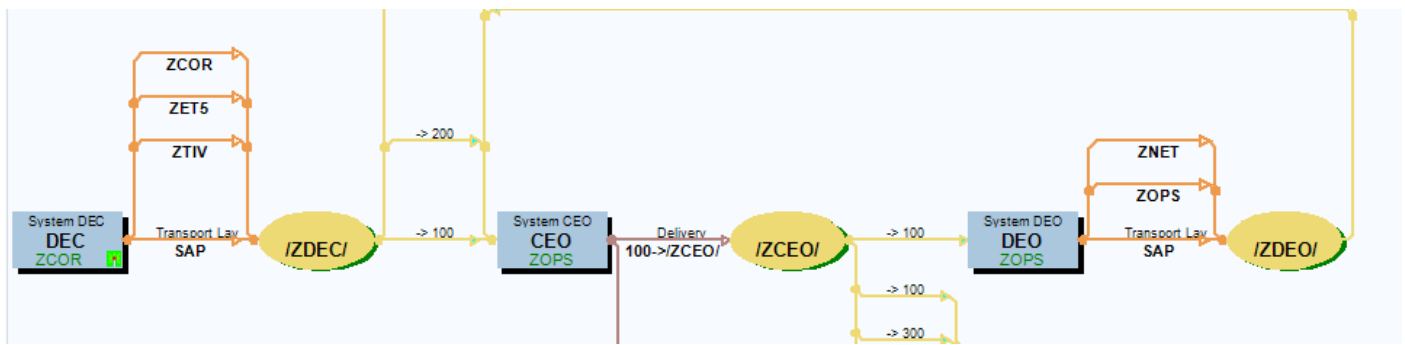
Import requests are to be addressed to the IT development coordination team with Data-/Co-Files, documentation (if available), Solution Manager work item number and target development system/client. They will organize the import of the transports into the initial system. Tests are to be performed and documented in the same way as for own developments. The testing and documentation tasks are in the responsibility of the requester (functional owner). "Documentation on work item" includes the storage of the confirmation mails of manual distribution of transports from the basis team.

The Solution Manager – Focused Build Work Item process steps does not cover the original transport distribution from third party vendors. Due to that have the project teams aligned to a manual distribution process for the time being.

It reduces the solution manager work item to a documentation tool without transport distribution information. The requester must request the transports of his third party applications similar to the initial import also to the target systems. The IT development coordination team organizes the distribution together with the S4U infrastructure team and documents the execution in the work item (if provided by the requester).

Agreed mitigation (adjustment in process) to use the full functionality of Focused Build: In all cases, where the external provider offers the opportunity to collect his developments in packages with a Z-transport layer, it is planned to define the layer in the TMS configuration and give the option to assign the third party transports to the work items. This will be done only on single request. Such transport layer **must not** be used for other developments.

Example with external layer (ZTIV):



The same approach is used for the BKZ application using the transport layer ZET5.

2.6. QA tool usage and integration into the transport release

The former SAP ABAP ATC quality check routines have been replaced by the two new tools "On-Change Control" (OCC) and "Control for Code (C4C)" from the company Onapsis since 23.02.2024.

The Onapsis solution offers various possible uses across the entire system landscape. As an add-on in the development systems for the QA process, for remote scanning of QA and productive systems and integration into the SAP standard development tools for permanent scanning of developments by developers.

The automated QA assessment procedure is part of the ABAP development process in all oneERP development systems. The tests now take place against the regular Onapsis standard test cases as well as some of E.ON's own test cases. The priorities of the test cases are also predetermined by Onapsis.

2.6.1. Control for Code (C4C)

Control for Code (C4C) enables a holistic approach to securing all of the ABAP code on all of your systems, throughout your system landscape, no matter how many systems are involved.

The single developer is still responsible to perform an C4C check manually and correct all error before releasing the task to the transport. The developer has the same scan options during the implementation as under ATC/SCI. Also for the usage of C4C were screen menus in the different development transactions, like SE80, SE24, SE38, SE10 modified.

Transports can also be checked with transaction /n/vforge/fidman in each development system and brings the developer into the position to add comments for specific findings. But you must not forget to deactivate the filter for "no issue".

The transport release rules remain unchanged. No Prio. 1 or 2 Finding must exist for a transport.

Based on the complains from the /INCC/-development team about misusages of these objects - inclusive creation of dependencies - was decided to implement the special testcase CTC 9001: Check /INCC/-namespaces.

It is not allowed to use or integrate /INCC/ namespace objects in coding for other namespaces. This rule applies only for DEO, where the /INCC/-objects are located.

According to SAP defines the implementation of SAP GUI Controls risks for the executed function as soon as they will be embedded in FIORI or WEB UI usages. Therefore, they should not be used. To draw the developers' attention to this, a new test case is currently in development in C4C, which will lead to a warning during the ToC and transport release process.

2.6.2.Integration into the transport release in Solution Manager (OCC)

On-Change Control (OCC) is an add-on for Solution Manager ChaRM. It is integrated into the Change Request Management process through the Transport-Related Checks, Custom Check assignment block, where customer-specific checks can be deployed to scan the transport requests attached in Change Documents or Work Items or Defect Corrections. On-Change Control is also a completely decoupled and flexible approval framework that enables approvers and requesters to run release approval of findings, from one single view, across the whole ABAP Landscape.

2.6.3.Exemption process

The exemption process is slightly changed against the former one from ATC. It begins now in the CD, for the single transport and not in the finding manager of the managed system anymore.

Based on the request from the developer has the QA team two different exemption accept options:

- Accept the finding: **Symbol will change to Green.** The Workbenchtransport can be released. The import of this finding into production without mitigation will be possible. Only valid for the specific finding at original TR.

- Accept the finding only for ToC: **Symbol will change to Yellow.** The ToC can be released, BUT not the Workbenchtransport. The Import of the ToC into the C-/S-systems for functional tests will be possible. For later import into QA and P systems, finding at original TR must be mitigated, otherwise status change will be blocked.

The decision is made transparent to the developer in PSM, by using different colors for the approval symbol in the single document.



Exemption requested



Accepted for test (ToC release only)



Accepted



Rejected



Permanent exemption

For further details, please see the respective process and FAQ documentations from the QA team. In case of additional support is needed, please contact the mailbox: EBS SAP Request Technical QA SAP-request-technical-qa@eon.com

2.9. Naming Conventions

2.9.1. Source code format conventions

Coding style

- Delete obsolete parts of the code. Don't comment it out
- Attributes, methods, events and types must have a meaningful English name
- Classical Dynpro fields require global data objects in the function group. Keep those global objects at an absolute minimum. Declare a dedicated structure for every screen - do not 'share' one global object with different screens.
- The software architecture should follow the MVC-principle: Do not mix coding that manages screen flow or appearance (as well as event handlers for screen elements) with screen-independent or business logic.
- Use always Pretty Printer
- Use classes instead of using form routines.

Inline comments

- All comments must be in English.
- Description of all objects with report source must be filled in the header.

Documentations

- In case of SAP standard code has been copied, customer changes must be commented and specific function module documented
- It is possible to set the short text as documentation for a data element (documentation status) or alternatively provide a proper documentation.
- A meaningful brief description must be created for each Dynpro.

Usability

- The SPRO path must be updated along with the brief description of the function. When linking the created Z-Function module in IMG, please enter a proper short description.
- Avoid list reports. Use SALV where available.
- To each field (parameter or select options) that appears in the selection screen, a corresponding identification should be assigned in the text elements.
- The look & feel of screen dialogs must comply with the SAP standard design.
- Use a standard GUI status or create a new one for every program. A GUI title is also obligatory.
- At the end of a process there must be a feedback for the user about the application state. It can be a LOG, an ALV list or a success/error message.
- Ensure that the user always finds a way to cancel processing and leave the dynpro quickly (e.g. the user shouldn't be forced to fill all fields correctly to finally press the 'Cancel'-Button).
- Using classical dynpro screens usually does not offer SAP standard access to the user documentation like in reports. In this case create a documentation in SE61 and display it with help of a button

Further Recommendations

- Internal tables with high number of rows should be cleared after usage.
- A table must be sorted before a READ TABLE BINARY SEARCH.
- "ORDER BY or sorting is necessary if the internal table is used in:
 - READ
 - CONCATENATE ... LINES OF itab ...
 - DELETE ... FROM/TO or INDEX
 - FIND FIRST OCCURRENCE ... IN TABLE
 - LOOP AT itab.
 - · AT ... ENDAT.
 - · EXIT/RETURN/LEAVE ...
 - · FROM/TO
 - MODIFY ... INDEX
 - PROVIDE ...
 - READ TABLE ... INDEX ...
 - REPLACE FIRST OCCURRENCE ... IN TABLE
 - SEARCH ... IN"
- Use whitelists or RegEx for external data if possible. At times, it is easy to limit user input to a limited set of valid values in a whitelist. Methods CHECK_WHITELIST_STR or CHECK_WHITELIST_TAB of class CL_ABAP_DYN_PRG may be used for this purpose.
- "Cooperative response to SAP locks: Before any changes are made to the database, the object to be modified in the LUW must be locked by means of the logical locking mechanism such that concurrent access by another or the same program is rendered impossible. In other words, when changing SAP

standard tables, the correct lock object should be determined first and used accordingly. In exceptional cases (e.g. XPRA) you can do without the lock object because concurrent access will be prevented in another way. In the exception must the developer explain (documentation included in program code) why locks are not necessary

- It is necessary to fill the secondary tables at the lock object if you want to change the data within logical connected tables, in this case it is not enough to lock only the header table.
- Directly changing global data from a form routine without parameters is forbidden (CHANGING). Exceptions must be described in the documentation of the form routine; variables must be listed in the header of the form routine.
- Generated user exits need to retain the interface.
- Includes where the Form routines are, must have the same module name as the FORM.
- Transport to all relevant clients, exception for references
- Implementation of subroutines in function modules are forbidden. The implementation must be placed in the proper include (e.g. ...F01).
- Customer (customizing) tables and views are client dependent. First column must be the MANDT or CLIENT as the field and MANDT as the data element.
- In Customizing tables must the "LOG data changes" checkbox be checked
- "Display/Maintenance allowed" is only allowed in connection with a generated table maintenance view. For customizing tables must the standard record entry routine be selected as well.

2.9.2.Naming conventions for repository objects

Naming conventions significantly enhance the readability of programs and therefore the quick understanding of developments.

In addition to the fixed conventions with respect to the prefix of a development object, the following general rules apply:

- Use meaningful and descriptive **English names**.
- Make use of the **maximum** allowable length of names.
- Use a maximum of 15 characters for the name of **tables** (because of the lock object name).
- Compound names must be separated by an **underscore**.
- If there is a standard **SAP or otherwise conventional** name, give preference to that name instead of your own invention (Use SAP transaction 'SAPTERM' to find out the common naming/ description where possible.)
- Use the following characters only: 'A-Z', '0-9' and '_' (no hyphens).
- Names **must not begin** with a number (0..9).
- The names of normal development objects in transportable development classes **must not end in "_D"**. Objects with this suffix are reserved for temporary backups of deleted objects.
- When using **tools and wizards** (e.g. Easy Enhancement Workbench), take naming conventions into account as much as possible

- All field labels must be filled by adequate descriptions. Length of field label is recommended to be set to the maximum character length: short = 10, medium = 20, long = 40, Heading= 55.
- Define the short text as documentation for a data element (documentation status) or alternatively provide a proper documentation.
- Application Component must be filled
- If you have too many objects for one package then use sub-packages for modularization.
- For a customizing table a table maintenance generator must be created either directly or through a maintenance view, and set the maintenance to allowed, where you create it.
- As module name is also the fictive value "Z_" (prospectively to be moved to the cloud), "ZE" (E.DIS only), "ZW" (Westenergie only) and "ZV" (VSE only; e.g. ZVHA for "Hausanschlüsse (house connection)") accepted. It will be used by the EAM project and only in the/ETN/-namespace. All developments to be created in these namespaces need a written confirmation from the respective Solution Architect and backend Architect (Thomas Kubiak and Andreas Bock) before the development is allowed to start.
- All developments for multi resource scheduling (MRS or MRSS) have to belong to the namespace: /ETN/PM_MRS.
- Fields in include structures: Especially for usage with suffixes. In case a structure needs to be extended by an include structure is the maximum length of a field name limited to 27 characters. This is necessary because includes can be defined using a max. 3 digits suffix and the complete fieldname must not be longer than 30 digits.

Object	Rule	Example /Comment
Authorization object class	Z<Namespace>	ZEON / ZKAW / ZSSF
Authorization object	Z<Characters>	
BAdi implementation class	<Namespace>CL_IM_<Module>_<Characters>	
BAPI Function module	<Namespace>BAPI_<Business object>_<Method>	
BOR object (type)	<Namespace><Characters>	Length 10 digits
Business object	<Namespace>BUS<Characters>	
Class	<Namespace>CL_<Module>_<Characters>	
Class Exception classes	<Namespace>CX_<Module>_<Characters>	
Class Interface	<Namespace>IF_<Module>_<Characters>	
CMOD	<Namespace><Characters>	
Domain	<Namespace><Module>D_<Characters>	
Data element	<Namespace><Module>D_<Characters>	
Enhancement implementation	<Namespace><Module>_<Characters>_I<2Digits>	
Enhancement implementation include	<Enhancement implementation>_<2Digits>	
Enhancement Point	<Namespace><Module>_<Characters>_P<2Digits>	
Enhancement Section	<Namespace><Module>_<Characters>_SC<2Digits> >	

Enhancement Spot	<Namespace><Module>_<Characters>_S<2Digits>	
Form: Adobe forms	<Namespace><Module>AF_<Characters>	
Form: Adobe forms interface	<Namespace><Module>AIF_<Characters>	
Form: SAPscript Form class	<Namespace><Module>FC_<Characters>	
Form: SAPscript Form name	<Namespace><Module>SS_<Characters>	
Form: SAPscript Standard text	<Namespace><Module>ST_<Characters>	
Form: SAPscript Style name	<Namespace><Module>SN_<Characters>	
Form: Smart forms	<Namespace><Module>SF_<Characters>	
Form: Smartforms Style name	<Namespace><Module>SSN_<Characters>	
Form: Smartforms text module	<Namespace><Module>STM_<Characters>	
Function group Class include	L<Function group>C <2Digits>	
Function group Event include	L<Function group>E <2Digits>	
Function group Form routines include	L<Function group>F<2Digits>	
Function group Input module include	L<Function group>I<2Digits>	
Function group Output module include	L<Function group>O<2Digits>	
Function group TOP include	L<Function group>TOP	
Function Group for WF	<Namespace><Characters>	Root cause is that exception the field length for these groups is limited to 10 digits.
Function Module	<Namespace>_<Module>_<Characters> Also accepted is: <Namespace><Module>_<Characters>	
Lock object	By Z tables: E<Z table name> By Partner Namespace: /partnernamespace/E<remainder Characters of Table name> By SAP Tables: EZ<SAP table name>	
Method	<Verb>_<Characters> <Verb> := get, set, calculate, etc. Event-Methodes: ON_<Characters>	
Message class	<Package>_<optional characters> also possible: <Namespace><Module>_MC<Characters>	
Number Range Object	<Namespace><Module>_<Characters>	
Query	<Namespace><Module>Q_<Characters>	
Query - User group	<Namespace><Module>UG_<Characters>	

Query – InfoSet	<Namespace><Module>IS_<Characters>	
Search help	<Namespace><Module>H_<Characters>	
Structure	<Namespace><Module>S_<Characters>	
Strukturen Append structure	<Structure/Table>_A<Number> by standard tables: <NS><module>S_<sap.table.name>_A<Number> or if required CI_<Structure/Table>	CI_Includes are to be defined as part of the corporate core in DEC, as the impacts to core tables cannot be excluded. See also the related Sub-Chapter "Table and structure Enhancements"
Fields in Append structures	The technical names for new components used in appends must start with Z or the individual namespace (preferred).	New 05/2021: ZZ seems to be rejected in HANA 2020 systems with message DT284
Table: Customizing table	<Namespace><Module>C_<Characters>	To be defined client dependent
Table: other tables (e.g.: Application Table)	<Namespace><Module>T_<Characters>	
Table: Index	Z<Digit> for SAP standard tables, otherwise free to decide	
Table type	<Namespace><Module>B_<Character>	
Transaction	<Namespace><Module>_<Character>	
Type group	The object name may be selected freely. Type groups are obsolete	
View	<Namespace><Module>V_<Characters>	Have to be defined with Authorization Group
Viewcluster	<Namespace><Module>VC_<Characters>	Have to be defined with Authorization Group
Workflow	See document "Development for SAP Business Workflow"	
Separately not listed objects (e.g.: Reports, OData Services etc.)	<Namespace><Module>_<Characters>	

View 4 Naming conventions for objects.

The names of the modules can be found in the transaction HIER in each SAP-System.

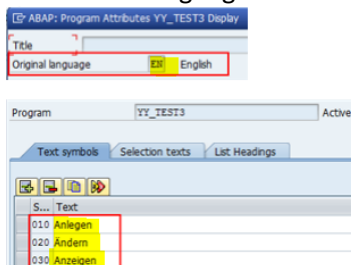
Enhancements are to be confirmed by the S4U development coordination and added always at the end of an existing DDIC table/structure definition.

For the definition of own fields is the usage of SAP standard data elements like AUFNR or CO_BELNR forbidden, as the report-report-interface (Berichts-Berichts-Schnittstelle) is not able to deal with it (selection values are being lost and calls ends up with wrong lists or dumps). Own data elements are to be used instead.

2.9.3. Translation related rules

During the installation and connection of the SAP Translation Hub (STH) some general advice for developments were judged as to be included.

- Never program hard coded values, because they would not be found during the translation process.
- Do not split translation relevant texts, as this could create wrongly translated values in other languages. (different sentence structures with information loss possible)
- Create consistent text entries. Take care to use the corporate and/or SAP terminology.
- Use a corporate guideline for entering texts (capitalization, terminology, formal/informal speech, abbreviations etc.).
- Some generic hints:
 - Don't use abbreviations. If you cannot avoid them, be consistent with the usage (e.g. Number should be always abbreviated as No., not Nr.).
 - Leave enough space for the texts. Example: if the German word 'Ja' has only 2 characters, you cannot translate it to 'Yes' with 3 characters.
 - Use numberings for variables. This helps the translators to keep the right ordering of the words. Example: Material &1 does not exist in order &2
- Do not copy English or German texts into texts in local language. Existing texts in local languages won't be overwritten by STH.
- Original language of the object and language of the entered text should always match.
 - Keep consistency between the original language of the object and the language of the texts created.
 - Do not mix languages within one object.



- Use language fields for texts in tables
 - The translation environment of the system recognize only tables with language fields (LANG or SPRAS) as translatable.

2.10. FIORI APP programming

Pre-Warning based on experiences from patching project in 2024: SAP no longer guarantees the backward compatibility of its FIORI apps as it used to. Under HANA, outdated application versions may be deleted again. However, if some of these have been reused in the customer's CLONE applications, this can lead to these applications no longer being able to run after the upgrade. What to consider when developing clones.

Within the S/4 HANA system environment FIORI App components become more and more important. To provide a consistent layout, control behaviour, and visual design in all SAP Fiori apps, we recommend the developer to use the SAP standard SAPUI5 front-end library as much as possible. E.ON own enhancements are allowed as soon as the SAP standard does not provide sufficient solutions.

But also the E.ON own enhancements have to follow the same rules and guidelines, like use the SAP Fiori launchpad, follow the described design rules, use predefined page layouts (with header, overview and details areas and footer), use the Analysis Path Framework or SAP Smart Business Framework, use SAP standard navigation and connection functionalities, consider SAP naming conventions (pages 12 ff), follow SAP JAVA coding guidelines (pages 14 ff). Consider also the points described under the chapter "Best Practices" (pages 32 ff). The referenced recommendation file is from SAP and called "SAP Fiori SAPUI5 Dev. Guidelines v3".



SAP Fiori SAPUI5
Dev. Guidelines v3.p

The development package names and transport layer definition follow the general rules of this document.

Extensions using the Fiori App "Custom Fields & Logic" are only permitted, if the economic benefit for E.ON is proven in writing by the requester. This has to be sent to the mailbox SAP ERP Solutions – Special Task Development SAP_ERP_Solutions_Special_Task_Development@eon.com for confirmation, before implementation. All adjustments in the ABAP systems must be summarized via an associated change document in PSM. This is also needed for the transport application.

2.11. Development Packages and Transport Layer for FIORI

For the distribution of developments is the assignment to a development package essential. Packages are generally names as follows: Name = /NS/XXPP

NS	Namespace
XX	Application component or module
PP	Free definition

The following central S/4 United transport layer are defined

DEC (core)	=>	ZCOR
DEA (core – add on)	=>	ZADD
DEO (operations)	=>	ZOPS
DEO (network/grid)	=>	ZNET
DET (transfer system)	=>	ZUWE
YEO (operations, network/grid)	=>	ZUWE

The naming convention for the catalogues themselves is defined as follows: /ID/YC.XXX.ZZZ

/ID/	= /System ID (+1(2))/
Y	= T(echnical) or B(usiness)
C	= Catalogue
.	full stop (separator)
XXX	= Module (e.g. FAP, FAR, FAA, PUR, SD_, MDG).
.	Full stop (separator)
ZZZ	Free text (optional)

Example: /EA/TC.MDG. (technical catalogue for master data governments in system DEA)

The **technical** catalogues are to be define in the Configuration Layer (client independent)

The **business** catalogues are to be define in the Customizing Layer (client dependent).

Also possible to separate new catalogues from old ones is the naming convention

/NS/	= /namespace (+1(3))/
Y	= T(echnical) or B(usiness)
C	= Catalogue

.	full stop (separator)
NDEV	New DEV catalogue
.	full stop (separator)
XX(X)	= Module (2-3 digits; e.g. FAP, FAR, FAA, PUR, SD_, MDG).
.	Full stop (separator)
ZZZ	Free text (optional)

Example: /EON/TC.NDEV.CA (technical catalogue for cross application data from system DEC).

Accepted exceptions for BKZ and ETS catalogue names are the pseudo modules:
/EON/*.NDEV.FBK* and /EON/*.NDEV.FET*.

Bring special attention to and describe in more detail within your technical specification document, the enhancements you have implemented (like In-App Extensions, Side-by-Side Extensions, Customized Extensions, View Extensions, Extension Points, Controller Extension; pages 36 ff).

Consider the Performance and Caching recommendations as well as the ones for the different security aspects phrased by SAP. System, Authorization and Role topics are not part of this development guideline.

Tests of your application are recommended to be performed on the browsers that are officially used within E.ON (like IE11, Edge, Chrome), because the implementation differs depending on the browser used.

For more details see the attached SAP Fiori SAPUI5 development guideline version 3.



SAP Fiori SAPUI5
Dev. Guidelines v3.p

And rules as part of the How to create custom tiles for Fiori Launchpad.



27032020-IT-S4U-E
ON_Fiori_Custom_Ti

2.12.S4U - Naming Conventions for embedded BI/BW

The whole BI/BW development guideline is stored in separate documents (one EXCEL and one WORD document, with same names) in Teams, but also included:



s4u - Naming
Conventions BW-BI



s4u - Naming
Conventions BW-BI

It is integrated part of this Guideline but has in individual update cycle.

Some extracts from the documents:

SAP e-on S4U Naming Convention for BI / BW										*) Length including the 5 characters for /EON/												
Abbreviations	MM=Main Module or Application component (1st level)										GG=Creators Group (AD=Support/Standard), PU=poweruser)											
	AP= Sub Module or Application component (2nd level)										T=Variable Type (V=Characteristic Value; H=Hierarchy; N=Hierarchy Node; F=Formula; T=Text)											
	ttt=Free text										P=Processing Type (M=Manual Input, R=Replacement Path, A=Authorization, C=Customer Exit)											
	nnn=cont. numbering										R=Select Option (S=Single Value; M=Multiple Single Value; I=Interval; O=Sel. Option; P=Preval. Value Set)											
C=Customizing (Y=Template, Z=Specific)										I=Type (O=Optional; M=Mandatory; I=Mandatory, Initial value not allowed)												
Area	Object Type	Max. Length *)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18...30	Example Technical Name	
System	Namespaces												Transport layer									
	(D)C(Q)P)EC	5	/	E	O	N	/					Z	C	O	R						/EON/	
	(D)C(Q)P)EA	5	/	E	Z	W	/					Z	A	D	D						/EZWI	
	(D)C(Q)P)EO	5	/	E	Z	O	/					Z	O	P	S						/EZO/	
(D)C(Q)P)EN	5	/	E	T	N	/						Z	N	E	T						/ETNI/	
Extractors	Package	30	/	E	O	N	/	M	M	A	P		t	t	t	t	t	t	t	t	t	/EON/CABI_BWOBJ
	DataSource	30	/	E	O	N	/	M	M	A	P	D	S	t	t	t	t	t	t	t	t	/EON/COPADStext
	CDS-View Extraction (SQL)	16	/	E	O	N	/	M	M	A	P	S	Q	t	t	t	t	t				/EON/COPASQtext
	CDS-View Extraction (View)	30	/	E	O	N	/	M	M	A	P		V	t	t	t	t	t	t	t	t	/EON/CABI_Vtext
	CDS-View Extension (nur View)	30	/	E	O	N	/	E		M	M	A	P		V	t	t	t	t	t	t	/EON/E_COPA_Vtext
	CDS-View Reporting (Private SQL)	16	/	E	O	N	/	P	M	M	S	Q	t	t	t	t	t	t				/EON/PCOSQtext
	CDS-View Reporting (Private View)	30	/	E	O	N	/	P		M	M	A	P		V	t	t	t	t	t	t	/EON/P_COPA_Vtext
	CDS-View Reporting (Interface SQL)	16	/	E	O	N	/	I	M	M	S	Q	t	t	t	t	t	t				/EON/ICOSQtext
	CDS-View Reporting (Interface View)	30	/	E	O	N	/	I		M	M	A	P		V	t	t	t	t	t	t	/EON/I_COPA_Vtext
	CDS-View Reporting (Consumption SQL)	16	/	E	O	N	/	C	M	M	S	Q	t	t	t	t	t	t				/EON/CCOSQtext
	CDS-View Reporting (Consumption View)	30	/	E	O	N	/	C		M	M	A	P		V	t	t	t	t	t	t	/EON/C_COPA_Vtext
	CDS-View Extraction (Table Function)	31	/	E	O	N	/	T	t	t	t	t	t	t	t	t	t	t	t	t	t	/EON/Ttext
	Customizing (Tabellen/Views) Template		Y	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	
	Customizing (Tabellen/Views) Lokal		Z	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	t	
InfoObject	InfoObject Business Content		aus Bus.Cont. übernommen																			
	InfoObject Characteristic	9	R	M	M	M	t	t	t	n	n											RCOCCOM01
	InfoObject KeyFigure	9	R	M	M	K	t	t	t	n	n											RCOKCOM01
	InfoObject Unit of Measurement	9	R	M	M	U	t	t	t	n	n											

SAP e-on S4U Naming Convention for BI / BW			*) Length including the 5 characters for /EON/																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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2.13.Naming conventions for data declarations enhancements

- **Global Data:** it is "visible" everywhere in the main object.
- **Local Data:** it is only "visible" in the form routines, methods or in the function modules where you declared it, nowhere else.
- **Attributes, types and aliases** of classes must be named as global data/constants/types.
- If you create an object which is called through SAP customizing, it **must have the same interface** like the sample. (e.g.: FI-CA Event sample function modules).
- Use the following characters only: 'A-Z', '0-9' and '_' (no hyphens)

Every data objects must have the pattern:

- <Data object> := <Scope><Type>_<Characters>

- **<scope> := (G | L | I | E | C | R | T)**
- **<type> := (C | V | S | T | R)**
- Every scope letter is attached to a meaning according to the table below:
- **G:** global object
- **L:** local object
- **I:** using/importing parameter
- **E:** exporting parameter (not allowed for Form routines)
- **C:** changing parameter
- **R:** returning parameter (not allowed for Form routines)
- **T:** types

Every 'type' letter is attached to a meaning according to the table below:

- **C:** constant
- **V:** general data field
- **S:** structure or work area
- **T:** internal table
- **R:** object reference

The following naming conventions are defined for parameters and select options:

select option

<Data object> := S_<Character>

Parameter

<Data object> := P_<Character>

- The definition of field symbols must be based on the data definitions listed above.

2.14. Modifications and other forbidden Techniques

The programming of direct write access to SAP tables and implicit enhancements are forbidden to support the clean core approach from SAP.

What is meant by “Clean Core”? Clean means up-to-date, documented, unmodified, consistent, efficient and cloud compliant. Clean Core has 5 dimensions of guiding principles for resilient business processes and extensions: Business Processes, Extensibility, Data, Integration, and Operations. ABAP custom code is cloud compliant if it is based on ABAP RESTful ABAP Programming (RAP) model with ABAP Cloud language which is restricted to public objects only.

Also Modifications to SAP Objects are not allowed.

If the characteristics of an E.ON-Data Element using a SAP Standard-Domain are requested to be changed, the solution **must not be** to modify the characteristics of the used SAP Standard-Domain (e.g. change the length or make a NUMC a CHAR Domain).

The Developer must check and decide whether:

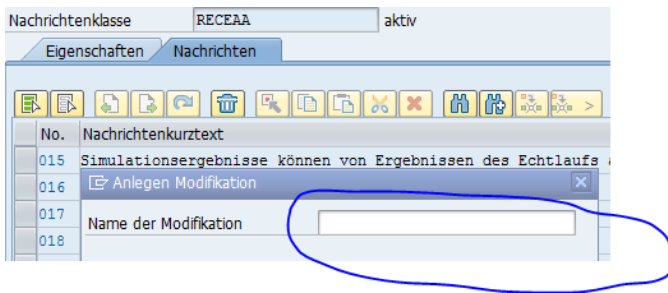
- Another SAP Standard Domain matches the new requirement (= replace the used Standard by another Standard Domain). SAP provides a lot of domains with different attributes. Transaction SE11, F4 value help for Domains:

SE-Repository Info System: Domains Find (62 Hits)		SE-Repository Info System: Domains Find (200 Hits)	
Domain name	Short Description	Domain name	Short Description
NUMC	Defined for data element NUMC: Test	CHAR	Character field length 2
NUMC_00_20	Two-Digit Number with Check on Interval 0 - 20	CHAR_01	One-Character Text Field, Any Contents
NUMC_00_20_IN	Two-Digit Number with Check on Interval 0 - 20	CHAR_0241	Character 241
NUMC_1_999	Commit work range 1-999	CHAR_122	122 character field length without checking
NUMC_50	50-digit	CHAR_124	Character field length 124 without check
NUMC_LG_05	Numeric field of length 5, without check	CHAR_132	Character Field, Length 132 Without Check
NUMC01	Numerical field (length 1)	CHAR_1536	Char 1536
NUMC015	Numeric with 15 places	CHAR_225	Field string for 225 characters
NUMC02	2-digit numeric field without check	CHAR_25	Character field, 25 characters long
NUMC03	Numerical field (length 3)	CHAR_30	30 Characters
NUMC04	Numerical field (length 4)	CHAR_35	Text field 35
NUMC05	Numerical field (length 5)	CHAR_4	Four-Character Text Field Without Check
NUMC06	Numerical field (length 6)	CHAR_40	Character Field (Length 40)
NUMC07	Numerical field (length 7)	CHAR_5	5-characters long without check
NUMC08	Numerical field (length 8)	CHAR_50	Character String 50 Characters with Lowercase Letters
NUMC09	Numerical field (length 9)	CHAR_512	Character 512?
NUMC0TO3	Nume Fixed Values from 0 to 3	CHAR_55	Text field length 55
NUMC1	1-character numeric fields without check	CHAR_60	CATS: Character string
NUMC1_VK	Numerical one-character field	CHAR_60_KL	Character String 60 Characters with Lowercase Letters
NUMC10	Numerical field (length 10)	CHAR00001	1 place character
NUMC10_SECOND_KEY	Secondary Key for Tables	CHAR0001	Individual, user-defined character
		CHAR0001PCAL	Individual, user-defined character

- Or only a new E.ON Domain matches the requirement (= replace Standard by E.ON Domain)

If a Modification cannot be avoided, e.g. as it is part of a SAP note implementation, it has to follow the existing modification process.

One example for new messages defined in a SAP Note is:



These kinds of implementations are to be done in the original language reported back by the system. The message translation is a second step using transaction SE63. The name of the modification must follow the namespace guideline from this document. E.g. in DEC it has to get name, like /EON/NOTE2841856 Sprache DE. (/NS/NOTE#number Language). The assigned development packages belongs also the /EON/ namespace.

2.15. Table and structure Enhancements

2.15.1. Conventional or standard enhancement with CI_Includes

Table enhancements have to be realized as follows:

Any new field in a SAP database table or structure is to be created with reference to SAP-Standard or /EON/ data elements and domains.

Corporate core enhancements for new fields have to be covered in general append structures created in the /EON/-Namespace. No direct field definition in the database table. Especially not in SAP Standard tables.

Example from DEC/100 system:

Transparent Table **SKA1** Active

Short Description: G/L Account Master (Chart of Accounts)

Attributes | Delivery and Maintenance | Fields | Input Help/Check | Currency/Quantity Fields | Indexes

Search Built-In Type 30

Field	Key	Ini...	Data element	Data Type	Length	Deci...	Short Description
FUNC_AREA	☐	☐	FKBER	CHAR	16	0	Functional Area
GLACCOUNT_TYPE	☐	☐	GLACCOUNT_TYPE	CHAR	1	0	Type of a General Ledger Account
LAST_CHANGED_TS	☐	☐	TIMESTAMP	DEC	15	0	UTC Time Stamp in Short Form (YYYYMMDDhhmmss)
.APPEND	☐	☐	/EON/FIGL SKA1_EXT	STRU	0	0	FI: G/L Acc. Segment A (SKA1) Extension - Customer fields
.INCLUDE	☐	☐	/EON/FIGL SKA1_CUST_FLDS	STRU	0	0	FI: G/L Accounts Segment A (SKA1) customer fields
/EON/INC_TYPE	☐	☐	/EON/FIGL_INC_TYPE	CHAR	1	0	FI: Income Type
/EON/MOVE_TYPE	☐	☐	/EON/FIGL_MOVE_TYPE	CHAR	10	0	FI: Movement Type SET
/EON/PARTNERA	☐	☐	/EON/FIGL_PARTNERA	CHAR	1	0	FI: Partner Assignment
/EON/INVESTA	☐	☐	/EON/FIGL_INVESTA	CHAR	1	0	FI: Investee Assignment
/EON/MATURITY	☐	☐	/EON/FIGL_MATURITY	CHAR	3	0	FI: Maturity
/EON/TRAN_TYPE	☐	☐	/EON/FIGL_TRAN_TYPE	CHAR	2	0	FI: Transaction Type
/EON/EIGEN	☐	☐	EIGEN	CHAR	8	0	Cost element attributes

SAP SE11 dec INS

Additional fields in CI_Includes and appends in CI_Includes have to be implemented exclusively in the corporate core development system DEC/100.

Structure **CI_AUFG** Active

Short Description: Kundenfelder Auftrag

Attributes | Components | Input Help/Check | Currency/quantity fields

Built-In Type 1 / 22

Component	Typing Method	Component Type	Data Type	Length	Deci...	Coordinate	Short Description
EZAIL	Types	/EON/COD_AIL	CHAR	50	0	0	Asset description
EZASN	Types	/EON/COD_ASN	CHAR	12	0	0	Asset super number
.APPEND	Types	/EON/PMWO_FV_CI_EQUI_APP	000	0	0	0	Additional fields
/EON/INBDT	Types	/EON/PMO_INBDT	DATS	8	0	0	Start-up date
/EON/AUMNG	Types	/EON/PMO_AUMNG	QUAN	13	3	0	
/EON/AUMEI	Types	/EON/PMO_AUMEI	UNIT	3	0	0	
.APPEND	Types	/IGY/PMWO_RE_CI_AUFG_APP	000	0	0	0	
/IGY/DONT_CHECK	Types	/IGY/PMWO_RE_DONT_CHECK_IMMO	CHAR	1	0	0	
/IGY/RCV_BUKRS	Types	/IGY/PMWO_RE_RCV_BUKRS	CHAR	4	0	0	
/IGY/RCV_ACASTY	Types	/IGY/PMWO_RE_RCV_ACASTY	CHAR	1	0	0	

Original language German (DE), Display in English (EN) SAP 100 dec INS

Additional fields in direct Append Structures enhancements in the operations systems or the grid systems are to be implemented the same way, but in the individual development systems and namespaces (DEA -> /EZW/ \ DEO -> /EZO/ or /ETN/ or /KAW/ or /INCC/).

Example from DEO with comparison to DEC

Transient Table **AUFK** Active

Short Description Order master data

Attributes Delivery and Maintenance Fields Input Help/Check Currency/Quantity Fields Indexes

Field	Key	In.	Data element	Data Type	Length	Dec.	Coordinate	Short Description
ERSETZ	☐		CO_BES_TIME	TIMS	6	0		0 Time created
ARETZT	☐		CO_CPS_TIME	TIMS	6	0		0 Changed At
CSTD_WTIME	☐		CV_WJVM	CHAR	4	0		0 Costing Variant
COSTESTR	☐		CV_ICAR	WSTR	12	0		0 Cost Estimate Number for Cost Est. w/o Q
VERBA_USER	☐		VERBA_USER	CHAR	12	0		0 Person Responsible for CD Internal Order
INCLUDE	☐		INCL_PPW_AUFK_P5	STR0	0	0		0 Order Master Data - Persistence Include
INCLUDE	☐		GVV_DATA_4	STR0	0	0		0 VV data: (gvv_data_1)vv_sca
APPEND	☐		CDPP_PPW_AUFK_THRST	STR0	0	0		0 Time stamp for created on and changed on
APPEND	☐		CDON_AUFK_AUFK	STR0	0	0		0 CJI specific APPEND for order master data
APPEND	☐		CDONACTY_A	STR0	0	0		
/NAM/EQUID	☐		/NAM/DE_EQUID	CHAR	16	0		0
/NAM/CONTRACT	☐		/NAM/DE_CONTRACT	CHAR	16	0		0
/NAM/CONTRACTCAT	☐		/NAM/DE_CONTRACTCAT	CHAR	16	0		0
/NAM/INTENRPRNT	☐		/NAM/DE_VTNG	CHAR	4	0		0
/NAM/VERTEILERSH.	☐		/NAM/DE_VTNG	CHAR	16	0		0
/NAM/ACTAUTPR	☐		/NAM/DE_ACTAUTPR	CHAR	15	0		0
APPEND	☐		INSSCT_APPEND_AUFK	STR0	0	0		0 TICH CHANGES
APPEND	☐		ADIAUFK	STR0	0	0		0 Append for AUFK

Transparent Table **ACFE** **Active**

Short Description **Order master data**

Attributes Delivery and Maintenance Fields Input Help/Check Currency/Quantity Fields Indexes

91 / 105

Field	Key InL	Data element	Data Type	Length	DecL	Coordinate	Short Description
EXEFTIME	☐	☐ CO_SIN_TIME	TIME	4	0		0 Time created
AEZEIT	☐	☐ CO_CMS_TIME	TIME	6	0		0 Changed At
CFG_VARIANT	☐	☐ CM_CFGVAR	CHAR	4	0		0 Coding Variant
COSECTOR	☐	☐ CM_SECTOR	NUMC	12	0		0 Cost Estimate Number for Cost Est. w/o Qty Structure
VERMA_USER	☐	☐ VERMA_USER	CHAR	12	0		0 Person Responsible for CO Internal Order
INCLUDE	☐	☐ INCL_NEW_AUFK_P2	STRTD	0	0		0 Order Master Data - Persistence Include
INCLUDE	☐	☐ INCL_DATA_4	STRTD	0	0		0 2V data: (gls_data_1)/gv_oco
APPEND	☐	☐ APPEND_AUFK_P1	STRTD	0	0		0 Time stamp for created on and changed on time
APPEND	☐	☐ APPEND_AUFK_P2	STRTD	0	0		0 CU specific APPEND for order master data
APPEND	☐	☐ APPEND_AUFK_P3	STRTD	0	0		0 TICH CHANGES
APPEND	☐	☐ APPEND_AUFK_P4	STRTD	0	0		0 Append for AUFK
APPEND	☐	☐ APPEND_AUFK_P5	STRTD	0	0		0 Work Center
APPEND	☐	☐ APPEND_AUFK_P6	STRTD	0	0		0 Append structure for Regulatory Reporting
APPEND	☐	☐ APPEND_AUFK_P7	STRTD	0	0		0 Network Leading Status
APPEND	☐	☐ APPEND_AUFK_P8	STRTD	0	0		0 Append for aufk

100 % dec INV

The only restriction is defined by SAP within the SAP Gateway Service builder. SAP is predefining for the schema namespace for the Data Model, that the customer field names must not contain special characters except "." and "_".

Original text from SAP:

Assigning Schema Namespace

To set Schema Namespace for projects of type **Service with SAP Annotations** and **Service with Vocabulary based annotations** proceed as follows:

1. Right click **Data Model** folder.
2. Select **Details**. **Details** dialogue box appears
3. In the field **Schema Namespace** enter schema namespace for the project. If this field is left blank, Schema Namespace is generated and populated after runtime generation based on Technical Name of the project.

i Note

Special characters (apart from '.' and '_') are not allowed. If entered, an error message is displayed.

Source: Schema Namespace - SAP Help Portal

That means, that our namespaces starting and containing "/" are not allowed for fields used in SAP Standard Fiori Apps, like My Timesheet. Fields beginning with "/" lead to the error "/EON/ZTART" is not a valid ID.

As workaround is allowed to name such fields with ZZ at the beginning.

Example from DEC CI CATSDB:

Component	Typing Method	Component Type	Data Type	Length	Deci...	Coordinate	Short Description
ZZREF_BELNR	Types	/EON/CATSD_REF	CHAR	32	0		0 Reference Document Timesheet
ZZDOC_CAT	Types	/EON/CATSD_DOC	CHAR	4	0		0 Document type
ZZREC_TYPE	Types	/EON/CATSD_REC	CHAR	6	0		0 Key figure Timesheet
ZZCATS_STATUS	Types	/EON/CATSD_CATS	CHAR	2	0		0 Confirmation status Cats (default times)
ZZREADINESS	Types	/EON/CATSD_READ	CHAR	1	0		0 Readiness

2.15.2. Enhancement with Custom Fields & Logic

The implementation of enhancements with Custom fields & logic is new with SAP HANA and regularly be used to enhance SAP standard tables with new customer own fields. In old releases this was realized with customer includes or append structures directly in the ABAP system. Custom Fields & Logic enhances the tables out of the FIORI APP.

The implementation with Custom Fields & Logic is more complex as in that case is the enhancement itself been performed outside the ABAP development system.

SAP itself is stating that: The management of data in an extension scenario deviates from the management of data in the standard scenarios. You are responsible for ensuring that the data used in an extension scenario is managed in accordance with any applicable legal requirements or business needs, such as data protection legislation or data life cycle requirements. Please note that the extensibility framework is currently not integrated in the privacy-by-default functionality. Therefore, the extensibility framework should not be used for the processing of personal data if this processing falls under any applicable data protection legislation. [Source](#)

Therefore it is only allowed after consultation of and approval from the SAP ERP Solutions – Special Task Development SAP_ERP_Solutions_Special_Task_Development@eon.com mailbox. For getting that, you need to provide a valid argumentation:

- why is the implementation with Custom Fields & Logic to be preferred?
- Which restrictions or other technical needs make it necessary to implement the enhancement with Custom Fields & Logic?
- which benefits will be realized with the implementation with Custom Fields & Logic?

The implementation will be reviewed a second time during QA with ATC, causing priority 1 + 2 findings and at least at this point in the development process, the developer must provide the above mentioned justification.

2.15.3. Indexes under SAP HANA

Table indexes are also created and required in a system environment based on an SAP HANA database.

We orient our usage on the latest version of the SAP Note 2160391 - FAQ: SAP HANA Indexes. Component: HAN-DB-ENG (SAP HANA DB Engines),

This note can be found also on our connect page: <https://connect.eon.com/files/5f4c5bca-4588-4fbc-a779-37bfa8a40362/7e65b3c0-1248-43b2-b3be-f979c7246ec2/2160391%20-%20FAQ%20SAP%20HANA%20Indexes%20pdf>

2.16. GDPR Enhancement

The GDPR (General Data Protection Regulation or DGSVO "Datenschutzgrundverordnung") was newly introduced in Europe in 2018. It describes the protection of the personal data of natural persons. Legal persons and organizations remain unaffected. According to Article 4 of the GDPR, personal identifiers include all information by which a natural person is identified or identifiable. Personal data is defined as all data that can be traced back to the master record of a person via technical numbers. (e.g. with help of an ABAP Query)

What are "personal data" (not complete)?

- Master data
- Address data
- Contact information
- Contract information
- Bank account data
- Invoices
- Rating information
- Creditworthiness
- Etc.

The GDPR related adjustments are required as soon as it cannot be guaranteed that only legal persons or organizations are being managed with the respective customer development. This applies for all S/4 United systems. This Guideline applies for the following namespaces /EON/*, /ETN/*, /EZO/, /EZW/, Z* and Y*.

2.16.1. Functional Requirement

After Implementation of the ILM (Information Lifecycle Management) functionality of simplified blocking, it is permitted to display or change blocked Customer / Vendor / Business Partner or Employee master data by end user. Only user with special authorizations remain in the position to display and maintain them as in the past. This statement is already correct for standard reports/function modules/classes, but not for custom own reports/function modules/classes.

All E.ON Customer own development objects need to be adapted to use the checks for blocked customer/vendor in same way, as this is implemented in standard SAP objects. To be differentiated are programs reading vendor-/customer-data one by one on one hand and Mass-reading-programs (the performance is to be kept in mind) on the other hand.

2.16.2. Technical Implementation

The next chapters will describe the technical implementation of the check in all S4United Systems for single master data record usage. As part of this S/4 Development guideline, it is to be considered by all developer working in any of the S4U systems.

The developer has the following options to prevent the customer own functionality from displaying blocked data records for unauthorized user:

- Initial check of the authorization of the executing user, directly after start-of-selection, store the return value in a global constant (like GC_BLACK (CHAR 1). Example values can be Blank (user is not authorized to see blocked records) or X (user has special authorizations to see all selected records) and consider this value in the further program coding (e.g. for DB table selects, PBO of screens or during output list preparation). See also the description for mass processing.
- During database table selections: Execute one of the following BAPI's, dependent from the database table to be read.
- Suppress the display of blocked records during output list preparation (in a separate PBO module for screens)

Whenever and Wherever one of the following vendor and customer master data tables need to be read or a select is required, one of the following BAPI's is to be executed before, this is obligatory : BAPI_BUPA_CENTRAL_GETDETAIL or BAPI_CUSTOMER_GETDETAIL2 or BAPI_VENDOR_GETDETAIL.

In other words, one of the BAPIs is to be called before a read/select statement for one following tables. An equivalent protection via other modules is also permissible.

The relevant Vendor Master Data tables are:

- LFA1
- LFB1

The relevant Customer Master Data tables are:

- KNA1
- KNB1
- KNVK
- KNVV

A special treatment is required for the Business Partner Master Data table

- BUT000

These are the different fields, which will be maintained by the block processes:

- LFA1-CVP_XBLCK -- Business Purpose Completed Flag
- LFB1-CVP_XBLCK_B -- Business Purpose Completed Flag
- KNA1-CVP_XBLCK -- Business Purpose Completed Flag
- KNB1-CVP_XBLCK_B -- Business Purpose Completed Flag
- KNVV-CVP_XBLCK_V -- Business Purpose Completed Flag
- KNVK-CVP_XBLCK_K -- Business Purpose Completed Flag
- BUT000-XPCPT -- Central Block for Business Partner

The additional check requirement needs to become part of the QA assessment process step, independent from the implemented technical solution (existing manual QA or SACI from virtual Forge or any other Third Party quality tool).

In case the functionality shall provide an information to the user, it can be implemented with the help of the message class "CVP_DP_ILM".

2.16.2.1.Vendor Data Protection

E.ON own developed objects need to be adapted to call function module BAPI_VENDOR_GETDETAIL before Vendor details are being read/selected from the database table. The relevant field, which describes the blocked vendor is the field LFB1-CVP_XBLCK_B (X = blocked; _ = not blocked).

The function module BAPI_VENDOR_GETDETAIL checks the lock indicator, together with the authorization of the user.

The differences are given back in the RETURN structure and the resulting table content. For a user that doesn't have the authorization or the vendor is blocked, a message is given back in the RETURN field and no BANKDETAIL are listed in Return tables.

Test Function Module: Result Screen	
Test for function group	LC01
Function module	BAPI_VENDOR_GETDETAIL
Uppercase/Lowercase	☐
Runtime:	107.619 Microseconds
RFC target sys:	
Import parameters	Value
VENDORNO	2098174
COMPANYCODE	9999
Export parameters	Value
GENERALDETAIL	2098174 Alena Skořodopolová
COMPANYDETAIL	2098174 9999
RETURN	000
Tables	Value
BANKDETAIL	0 Entries
VENDORIBANDETAIL	1 Entry
	0 Entries
	0 Entries

Test Function Module: Result Screen	
Test for function group	LC01
Function module	BAPI_VENDOR_GETDETAIL
Uppercase/Lowercase	☐
Runtime:	110.966 Microseconds
RFC target sys:	
Import parameters	Value
VENDORNO	2098177
COMPANYCODE	9999
Export parameters	Value
GENERALDETAIL	2098177 Alena Svobodová
COMPANYDETAIL	2098177 9999
RETURN	IFN 524Vendor 0002098177 exists, but there is no bank data
Tables	Value
BANKDETAIL	0 Entries
VENDORIBANDETAIL	0 Entries
	0 Entries
	0 Entries

Same from QEO/100 (my user has the authorization to see the bank details and the data are anonymized).

Test for function group	LC01
Function module	BAPI_VENDOR_GETDETAIL
Case-sensitive	☐
Runtime:	70.380 Microseconds
RFC Target	
Import parameters	Value
VENDORNO	42000750
COMPANYCODE	3007
Export parameters	Value
GENERALDETAIL	42000750 ALUCA GMBH
COMPANYDETAIL	42000750 3007
RETURN	000
Tables	Value
BANKDETAIL	0 Entries
VENDORIBANDETAIL	2 Entries
	0 Entries
	2 Entries

View FM test result screens.

2.16.2.2. Customer Data Protection

To enable functionality that only users with correct authorization role can see blocked customers, all E.ON own developed reports need to be adapted to call function module BAPI_CUSTOMER_GETDETAIL2.

The relevant field, which describes the blocked customer is the field KNB1-CVP_XBLCK_B (X = blocked; _ = not blocked).

Test for function group	LC01
Function module	BAPI_CUSTOMER_GETDETAIL2
Case-sensitive	☐
Runtime:	23.828 Microseconds
RFC Target	
Import parameters	Value
CUSTOMERNO	18025010
COMPANYCODE	3007
Export parameters	Value
CUSTOMERADDRESS	18025010 XXXXX XXXXX
CUSTOMERGENERALDETAIL	18025010 2297475 000000000000 0 23.07.2021
CUSTOMERCOMPANYDETAIL	18025010 3007E4 Y009
RETURN	IFN 505Customer 0018025010 exists, but there is no bank data
Tables	Value
CUSTOMERBANKDETAIL	0 Entries
Result:	0 Entries
CUSTOMERIBANDETAIL	0 Entries
Result:	0 Entries
CUSTOMERSEPADETAIL	0 Entries
Result:	0 Entries

View FM test selection screen.

The function module BAPI_CUSTOMER_GETDETAIL2 not only checks the lock indicator, but also the authorization of the user. The return value is given back with the RETURN table. If no authorization exists (or the customer does not exist), message type 'E' or 'A' is being given back.

Test Function Module: Result Screen



Test for function group LC01
Function module BAPI_CUSTOMER_GETDETAIL2
Uppercase/Lowercase ☐

Runtime: 54.944 Microseconds

RFC target sys:
[]

Import parameters	Value
CUSTOMERNO	8201222
COMPANYCODE	7600

Export parameters	Value
CUSTOMERADDRESS	 8201222 Unbekannter Einzahler
CUSTOMERGENERALDETAIL	 8201222 964849
CUSTOMERCOMPANYDETAIL	 8201222 7600 000 Y000
RETURN	 EFN 504You are not authorized to display this customer

Tables	Value
CUSTOMERBANKDETAIL	 0 Entries
Result:	 0 Entries
CUSTOMERIBANDETAIL	 0 Entries
Result:	 0 Entries

Test for function group LC01
Function module BAPI_CUSTOMER_GETDETAIL2
Uppercase/Lowercase ☐

Runtime: 14.623 Microseconds

RFC target sys:
[]

Import parameters	Value
CUSTOMERNO	8201222
COMPANYCODE	7600

Export parameters	Value
CUSTOMERADDRESS	 8201222 Unbekannter Einzahler
CUSTOMERGENERALDETAIL	 8201222 964849
CUSTOMERCOMPANYDETAIL	 8201222 7600 000 Y000
RETURN	

Tables	Value
CUSTOMERBANKDETAIL	 0 Entries
Result:	 1 Entry
CUSTOMERIBANDETAIL	 0 Entries
Result:	 1 Entry
CUSTOMERSEPADETAIL	 0 Entries
Result:	 0 Entries



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View FM test result screens (left side: user with limited rights, right side: user with multiple rights).

The differences for blocked and not blocked customer, returned by the BAPI was successfully evaluated with a test user, not having the authorization to see blocked master records. Find the differences in the above two list screens of the test. To be considered: All IT user have the authorization to see blocked user in the MU D/Y/Q systems.

Other data base tables which shall be read only considering the master data block logic are:

- KNVK Customer Master Contact Partner
- KNVV Customer Master Sales Data

2.16.2.3. Business Partner Data Protection

The BUT000 table combines Master data records for Vendor and Customer. Before the read or select statement is programmed in the coding, the developer is requested to implement the execution of the BAPI_BUPA_CENTRAL_GETDETAIL. The results (display and process further or not) need to be considered the same way as the results (return codes) for vendors and customers. The blocking indicator field in BUT000 is XPCPT (x = blocked; _ = not blocked). In case the user is not authorized to see the blocked business partner will the error message R11 002 be returned.

Structure Editor: Display RETURN from Entry 1

Column Entry Metadata

1 Entry

ID	NUM	MESSAGE
E R11	002	You are not authorized to display this data; see long text

SAP | QGK (1) 100 | qgk | INS

View The delivered error message from the BAPI.

The function module processes the authority check for the executing user and business partner master record, In the case the user has no sufficient rights and the business partner is blocked, it returns an error message in the structure "Return".

A possible example coding can look like:

```
CALL FUNCTION 'BAPI_BUPA_CENTRAL_GETDETAIL'
  EXPORTING
    businesspartner      = ls_partner-partner
  IMPORTING
    centraldataperson    = ls_centraldataperson
    centraldataorganization = ls_centraldataorganization
    return               = lt_return
```

IF return <> initial.

Check for above error message E R11 002 and list it.

In dialogue (PBO): Error message that the user is not allowed to display the blocked record

In other process variants: no data selection executed for this blocked record

ENDIF.

It is in the responsibility of the implementing developer to use the correct BAPI, based on the requirement described in the functional of technical design document.

2.16.2.4. Protection of data used in mass-processes

Mass-programs are characterized in the way, that they are dealing bigger amounts of master data records in one execution and performance of database read processes are very critical.

The usage of these kind of programs is to be protected with the help of user oriented checks (is the executing user (the name is available in the field SY-UNAME) authorized to display blocked records or not. In case he is authorized, no further check is required. In case he is not authorized, checks on record level are needed with the respective negative impacts to the performance.

Authorization can be checked of following way:

For Business Partner by checking authorization object B_BUP_PCPT activity 03

AUTHORITY-CHECK OBJECT 'B_BUP_PCPT'

ID 'ACTVT' FIELD '03'.

For Customer by checking authorization object F_KNA1_BED begru FIAU activity 03

AUTHORITY-CHECK OBJECT 'F_KNA1_BED'

ID 'BRGRU' FIELD iv_begru

ID 'ACTVT' FIELD iv_activity.

For Vendor by checking authorization object F_LFA1_BEK begru FIAU activity 03

AUTHORITY-CHECK OBJECT 'F_LFA1_BEK'

ID 'BRGRU' FIELD iv_begru

ID 'ACTVT' FIELD iv_activity.

Checking on record level should include querying of relevant master data table for blocking indicator,

When blocking indicator is set, records have not to be displayed for regular user.

In case the functionality shall provide an information to the user, it can be implemented with the help of the message class "CVP_DP_ILM".

Further hints:

- *As far as outbound interfaces are concerned, it is to be decided together with the other party, whether:*
 - *A) The block flag is to be handed over (export record structure definition to be enhanced with the field CVP_XBLCK_* < Business Purpose Completed Flag >?) or*
 - *B) No records for blocked partner can be transferred*

Per default both situations can occur. In scenario A) is the separation logic to be implemented on the target system side. In scenario B) is the separation logic to be implemented on the source/extraction system side.

What is the guidance on inbound interfaces that create vendor payments in case the vendor is blocked? This is to be clarified by the developer together with the functional requester individually per task.

How to deal with technical user (like RFC user or system user) in E.ON own programs and functions? Per default are technical user in GLOBE not allowed to see blocked data records, as they are not planned to get special authorizations.

OPEN: same in S/4 United systems?

2.17.Obsolete program statements

From SAP as obsolete defined program statements are to be avoided and substitutes should be used instead.

2.18.Unlimited usage of programming commands

The unlimited usage of the command WAIT UP TO xxx is forbidden, because it interrupts the data and transactional flow and it has therefore a negative impact to the performance.

2.19.Exits

2.19.1. Program Exit - Framework

Program exits deliver new and E.ON-individual functionalities within the SAP standard transactions. The standard framework for their development is defined by SAP and listed in the following picture.

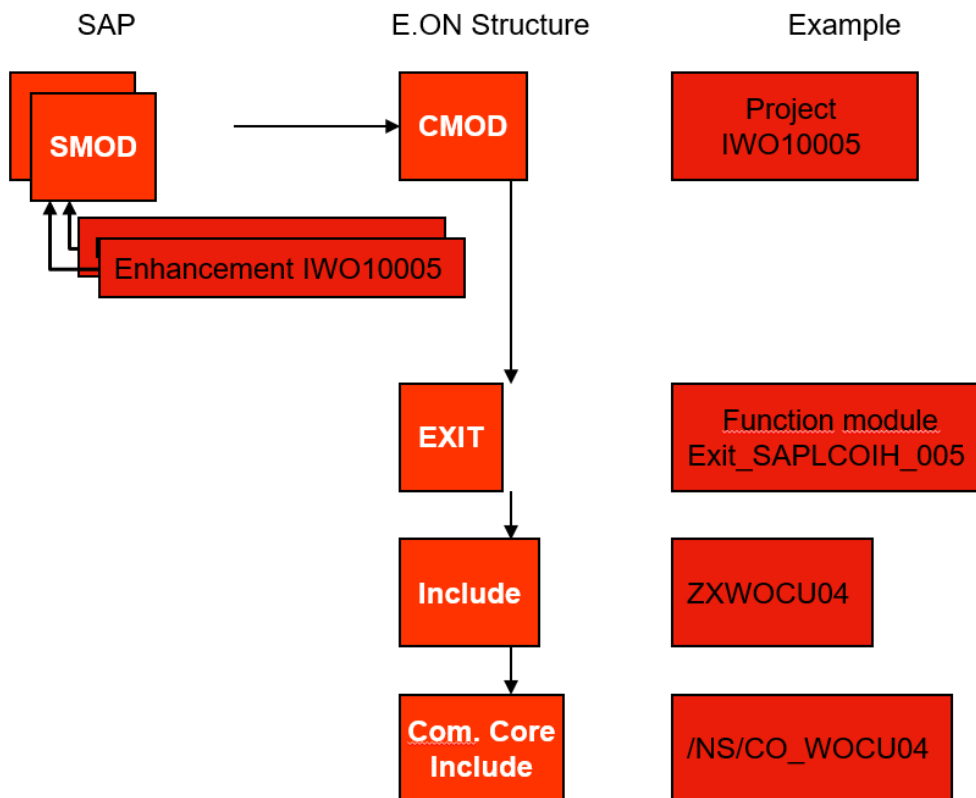


Figure: Exit development framework

The required exit (can be more than one function) can be collected from different SMOD Enhancements of the SAP. They have to be covered by one CMOD project. Each exit can be part of only one CMOD project. Existing assignments of exit components to one CMOD project are to be checked by the technical analyst before starting the development. It can be checked with the help of the table MODACT. Important are not only assignments in the template development system but in the market unit development system as well. The assignment has to be documented in the technical design. The CMOD project must have the same name as the corresponding SMOD enhancement.

Pure Scope user exits have to be developed manually in the core development system (DEC). They must be developed generically and controlled by a steering or control table. All objects belong to the /EON/ and ZX-namespaces (For CMOD-projects and ZX- includes allows SAP only the usage of Z-Packages, which is conform to this guideline). We strongly recommend the definition of one z-package per module, e.g. Z_MM_CM0D_EXITS for MM.

Pure Operations or Grid user exits have to be developed manually in the operations development system (example DEA or DEO). They must be developed generically and controlled by a steering table. All objects belong to the respective namespaces (/ETN/ or /EZO/ or /EZW/ or /KAW/ or /INCC/).

Shared exits are a mix of both. They contain a template framework, which means a template control table and a template enhancement class with a corresponding interface. In case that the central functions don't require the exit up to the requirement date of one market unit, the template class can even be empty. As long as the steering functionality for the shared exit can be fulfilled by the template control table, no market unit specific control table is necessary.

New/Additional lines of code **must not** be defined in the enhancement section itself. It must be encapsulated in own, new classes/methods or function modules. This is strongly recommended to keep the original coding readable for all developer and all business units.

Example from DEO:

The "Not allowed" program version, where the additional coding is implemented directly after the "Enhancement" statement.

```

549  ENHANCEMENT 1 /ETN/PM_GENERATE_EQUNR_TXT_I02. "active version
550
551  *%*****
552  *% This enhancement is implemented to automatically generate
553  *% Equipments short texts based on rule set in /etn/pmc_ds_rule
554  *%*****
555  *Modification Log:
556  *%-----
557  *% Version  User      Date      Reference      Transport Request
558  *%-----
559  *%
560  *% Initial  GS5076    13.09.2022    EAM-413      DEOK944703
561  *%*****
562
563  DATA: lv_eqctx TYPE ktx01.
564  DATA: lv_text   TYPE ktx01,
565         lv_field  TYPE ktx01,
566         lv_string TYPE string,
567         lv_separator TYPE char1.
568  CONSTANTS: lc_space TYPE char1 VALUE '~'.
569
570  * Begin of change S60548 24/11/2022
571  DATA: lv_tech_cat1 TYPE /etn/pmd_tech_obj_cat,
572         lv_active1   TYPE char1.
573  FIELD-SYMBOLS: <ls_itobl> TYPE itob.
574  CONSTANTS:
575     lc_tech_obj1 TYPE /etn/pmd_tech_obj VALUE 'EQ',
576     lc_enh_name1 TYPE /etn/pmd_enh_name VALUE 'SAPMIEQ0',
577     lc_imp_name1 TYPE /etn/pmd_imp_name VALUE '/ETN/PM_GENERATE_EQUNR_TXT_I02'.
578
579  ASSIGN ('(SAPMIEQ0)ITOB') TO <ls_itobl>.

```

The "Allowed" program version, where the additional coding is implemented in a separate Include. After the "Enhancement"-statement have we only one Include-statement:

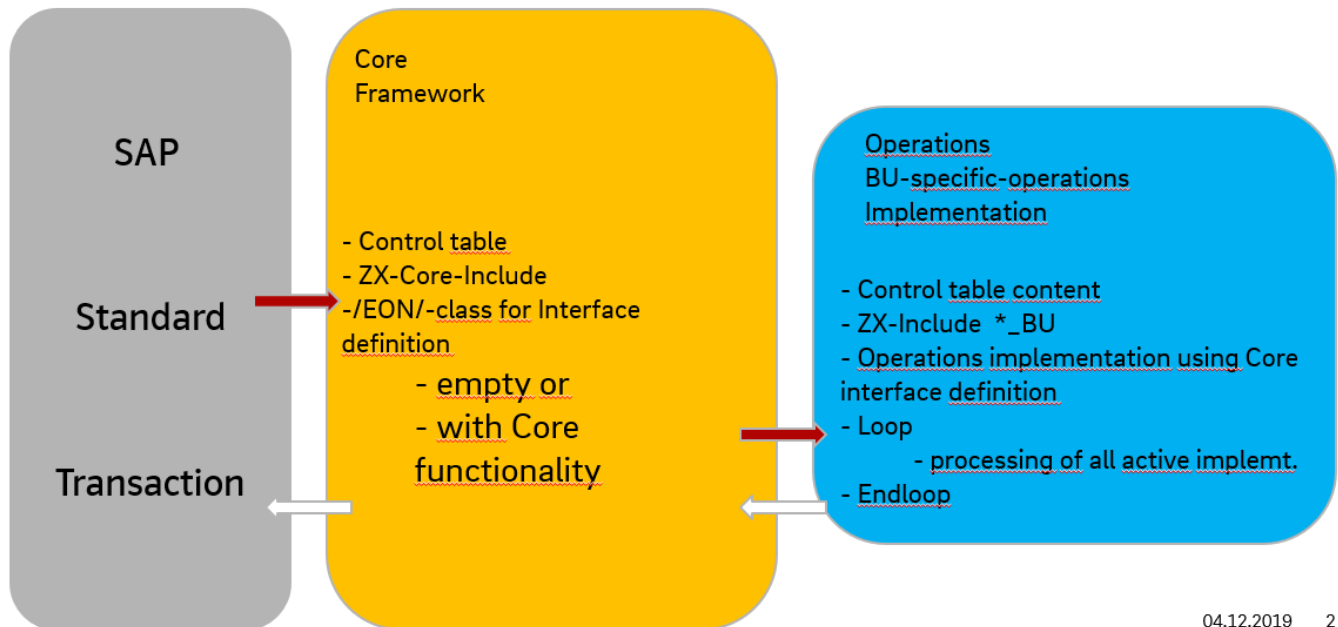
```
ABAP Editor: Include MIEQ0F69 anzeigen

Include MIEQ0F69 aktiv

549  ENHANCEMENT 1 /ETN/PM_GENERATE_EQUNR_TXT_I02.  "active version
550
551  *-----*
552  *& This enhancement is implemented to automatically generate
553  *& Equipments short texts based on rule set in /etn/pmc_ds_rule
554  *&-----*
555  *Modification Log:
556  *-----*
557  *& Version  User      Date      Reference      Transport Request
558  *-----*
559  *&
560  *& Initial  G25076    13.09.2022    EAM-413      DEOK944703
561  *&-----*
562
563  INCLUDE /ETN/PM_GENERATE_EQUNR_TXT_I01.
564
565
566
567  ENHENHANCEMENT.
568  ENHANCEMENT 1 /ETN/PM_UPDATE_CHNGE_CNTRL_I03.  "active version
569
570  *&-----*
571  *& This enhancement is implemented to save EQUIPMENT in change log table
572  *& /ETN/PMC_CHNGLOG table is updated for summation and SBL calculation
```

The process flow within the SAP standard transaction is illustrated in the following figure.

Process Flow



04.12.2019 2

Figure: Process flow

The selected namespace for CMOD customer projects created in all systems follow this development guideline.

2.19.2. Program Exit – in TOP-includes

In the template development system we define template relevant objects in a /EON/-top include for protecting this definition against changes in the market unit system. This helps to guarantee, that the template keeps running, although the non-scope area is specifically changed.

The common Top Include ZXxxxTOP has to remain free of individual coding.

For the specific requirements for one or more market unit, we create the possibility to develop an own include in the include Z*TOP_MU. We explain this with the example of an INCLUDE for PM: ZXQQMTOP. The include program ZXQQMTOP_MU does not exist in the Core development system (DEC), but can be implemented and enhanced in the operations development system (e.g. DEN or DEO) individually.

2.19.3. Program Exit – in other includes

The first step is to check whether the user exit is already implemented in the S/4 United Core system.

We consider 2 different cases with Operations and Core

2.19.3.1. User-Exit is a MU-specific Operations implementation

2.19.3.1.1. Namespace for objects created in the DEO system

Operations developments remain in the individual Namespaces and according to the actual E.ON Development Guidelines. They will be developed decentralized in the development system of the requiring operational unit.

In special cases operations-Developments are not migrated into the core namespace, but shall be distributed across the different operations systems. Therefore the maintenance must be centralized. For the distribution has the established ChaRM core process to be used. Example include is the HRUSER enhancement. In such case the developer or technical analyst must ensure that he uses only his own objects or the used existing objects fit to his development.

2.19.3.1.2. Namespace of objects in the ZX* Include

The ZX* Include in the function module-exit contains the operations include, which must be created in the operations namespace.

2.19.3.1.3. Coding in Operations-include

The coding has to be developed generically according to a pattern. This will be explained in the following chapters more in detail.

2.19.3.1.4. Steering table

The coding implementation must be controlled by a steering or control table.

- Generally one steering table must be created per function module. If multiple Exits depend on the same circumstances as far as the activation is concerned, one control table per CMOD is also accepted.
- An E.ON organization entity must be part of the steering table key, for example Company Code (BUKRS), Plant, Maintenance Plant (SWERK) or Controlling Area (this must be agreed with functional team). The implementation must be always clearly related to the respective value of this key.
- The used implementation class must be entered in each record. The same implementation class can be entered for several organization-entities (see view 11), if each organization-entity uses the same functionality. In the other case (different functionality per organization-entity) different implementation classes must be created. By maintaining the steering table, the local user can mark the respective class as active if he wants to use the code. They will be processed in the program one after another.
- Organization entity values in the steering table can be entered masked (Example for Plant 60* or 60++). This must be considered during programming.
- Control table entries must not be defined indefinitely or unlimited.
-

Transparent Table **/EON/MMC_XM06U36** Active

Short Description MM: customer exit MM06E005 exit 006 pbo custscr

Attributes Delivery and Maintenance Fields Input Help/Check Currency/Quantity Fields Indexes

Search Built-In Type 8

Field	Key	Ini...	Data element	Data Type	Length	Deci...	Short Description
MANDT	☒	☒	MANDT	CLNT	3		0 Client
BSTYP	☒	☒	BSTYP	CHAR	1		0 Purchasing Document Category
BSART	☒	☒	BSART	CHAR	4		0 Order Type (Purchasing)
EKORG	☒	☒	EKORG	CHAR	4		0 Purchasing organization
BUKRS	☒	☒	BUKRS	CHAR	4		0 Company Code
SEQNR	☒	☒	/EON/BCD_CM0D_S...	NUMC	4		0 Sequence number of class execution during CMOD processing
CLAZNAME	☐	☐	SEOCLSNAME	CHAR	30		0 Object Type Name
ACTIVE	☐	☐	/EON/BCD_CM0D_A...	CHAR	1		0 Active indicator of a class in a CMOD exit control table

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View Example of steering table (data dictionary)

Data Browser: Table /EON/MMC_XM06U36 Select Entries 1

Check Table...

MANDT	BSTYP	BSART	EKORG	BUKRS	SEQNR	CLAZNAME	ACTIVE
100	F	++++	++++	++++	0100	/EON/CL_MM_CM0D_EXIT_ZXM06U36	X

SAP SE11 dec INS

View Example of data in steering table

2.19.3.2. User-Exit is centrally implemented in the commercial core system

In some cases it can happen, that the same exits will be used in commercial core (DEC) and operations or grid functionalities (joint usage). In this case the processing and sequence of user exit implementations must be controlled by the steering table as well.

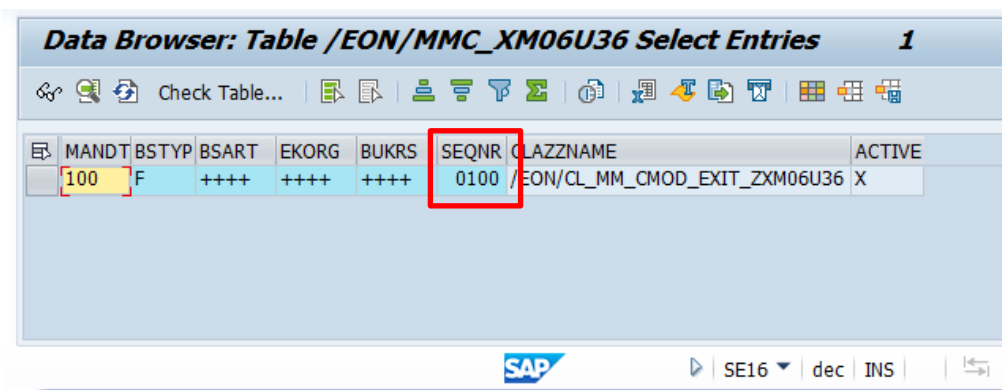
Commercial core changes have to be made in the DEC development system (after agreement with the functional team).

The /EON/* Include from the commercial core is already inside the generic ZX*-Include according to the pattern implemented. The operations-specific coding must be implemented within the /NS/-methods and includes. Due to the fact that **/EON/* Includes can't be changed in BU systems**, operations coding must be programmed in "process" method in the operations-NS-class and must be controlled by the same steering table. This helps to avoid conflicts.

The steering tables are already created in the Core system, so that they exist in BU systems, too.

Important: the field clazz_name or clazzname **must not** be defined as a key field, otherwise the MUs can create a new record with the same sequence, but with another class. So the main function of the field sequence will be bypassed. Every Template table entry has to have the highest sequence number as the template class has to be processed at last.

Example: Values in BU System



MANDT	BSTYP	BSART	EKORG	BUKRS	SEQNR	CLAZZNAME	ACTIVE
100	F	++++	++++	++++	0100	/EON/CL_MM_CMOD_EXIT_ZXM06U36	X

View 14 Example of data in Steering table

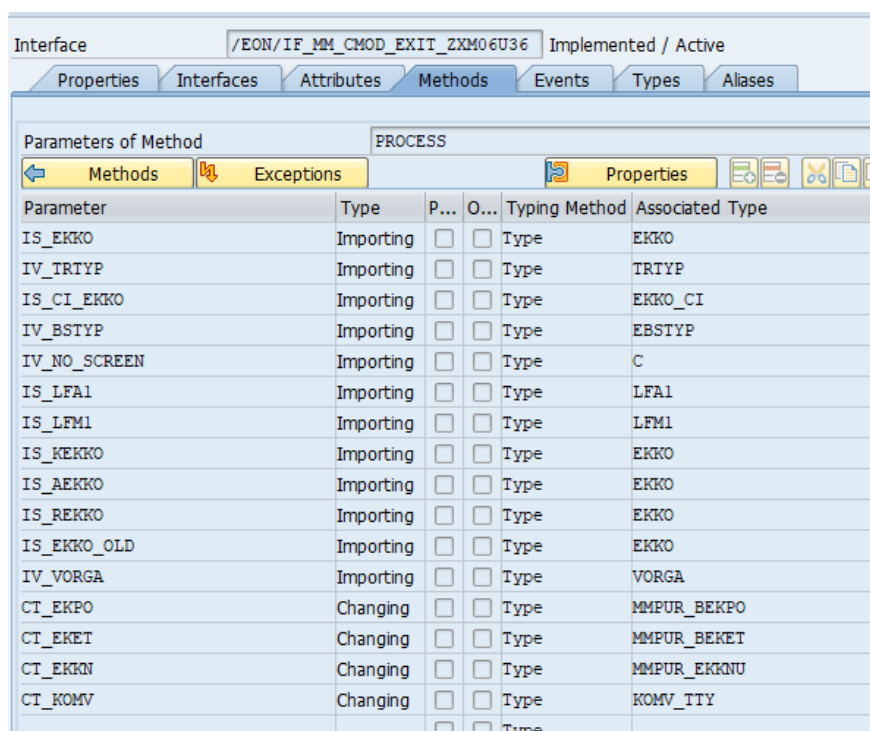
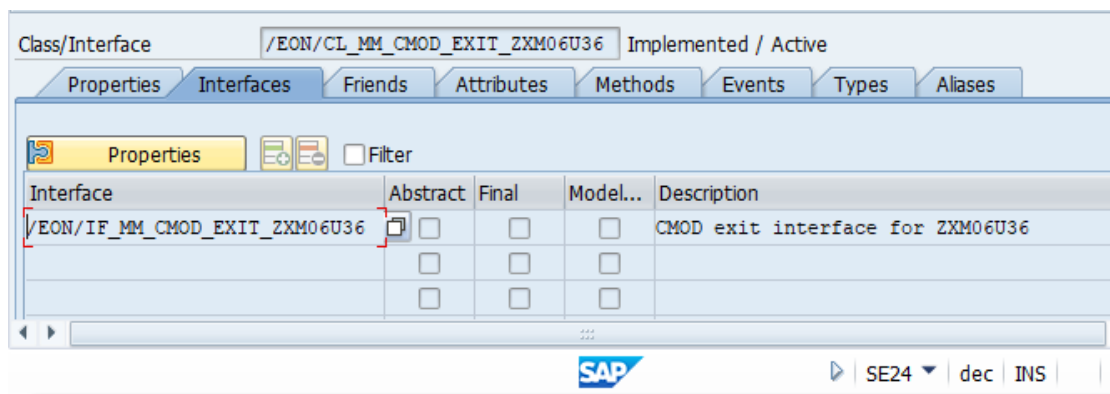
For each organization-entity all active implementation classes in the program's steering table will be selected.

The field SEQNR or SEQUENCE controls that the corresponding S/4 United operations implementation will be processed at first - as long as their sequence number is less than 100 - and the Core implementation will be processed at last. As long as this logic is considered, in one system, different business units can use different implementations in parallel.

The developer is responsible to take care that the called classes/functions, defined in the steering table have the same interface definition. This serves to ensure the hopeless execution of programming commands from the outset and to improve performance.

2.19.3.2.1. Interface definition.

All classes/sub-objects have the same interface definition.



```
Method /EON/IF_MM_CMOD_EXIT_ZXM06U36~PROCESS active

1 METHOD /eon/if_mm_cmod_exit_zxm06u36~process.
2
3 * ZINA
4 CALL FUNCTION '/EON/MM_ZINA_M06E_006'
5 EXPORTING
6     is_ekko      = is_ekko
7     iv_trtyp     = iv_trtyp
8     is_ci_ekko   = is_ci_ekko
9     iv_bstyp     = iv_bstyp
10    iv_no_screen = iv_no_screen
11    is_lfal      = is_lfal
12    is_lfm1      = is_lfm1
13    is_kekko     = is_kekko
14    is_aekko     = is_aekko
15    is_rekko     = is_rekko
16    is_ekko_old  = is_ekko_old
17    iv_vorga     = iv_vorga
18 CHANGING
19    ct_ekpo      = ct_ekpo
20    ct_eket      = ct_eket
21    ct_ekkn      = ct_ekkn
22    ct_komv      = ct_komv.
23
24 ENDMETHOD.
```

```
Function module /EON/MM_ZINA_M06E_006 Active
Attributes Import Export Changing Tables Exceptions Source code

1 FUNCTION /eon/mm_zina_m06e_006.
2
3 """Local Interface:
4 """ IMPORTING
5 """     VALUE(IS_EKKO) TYPE  EKKO
6 """     VALUE(IV_TRTYP) TYPE  TRTYP
7 """     VALUE(IS_CI_EKKO) TYPE  EKKO_CI
8 """     VALUE(IV_BSTYP) TYPE  EKKO-BSTYP
9 """     VALUE(IV_NO_SCREEN) TYPE  C
10 """     VALUE(IS_LFA1) TYPE  LFA1
11 """     VALUE(IS_LFM1) TYPE  LFM1
12 """     VALUE(IS_KEKKO) TYPE  EKKO
13 """     VALUE(IS_AEKKO) TYPE  EKKO
14 """     VALUE(IS_REKKO) TYPE  EKKO
15 """     VALUE(IS_EKKO_OLD) TYPE  EKKO OPTIONAL
16 """     VALUE(IV_VORGA) TYPE  VORGA
17 """ CHANGING
18 """     REFERENCE(CT_EKPO) TYPE  MMPUR_BEKPO
19 """     REFERENCE(CT_EKET) TYPE  MMPUR_BEKET
20 """     REFERENCE(CT_EKKN) TYPE  MMPUR_EKKNU
21 """     REFERENCE(CT_KOMV) TYPE  KOMV_TTY
22 """
```

View of Interfaces

The "process" method will be called in LOOP AT < internal_table > (the Control table with all existing and active implementations).

The developer must also take into account the content of the Core customizing tables. For this reason, a solution has been implemented which protects the Core table entries against changing them in the operations area.

2.19.3.2.2. Maintenance view authority check – Template Protection

A standalone authority solution has been created, to avoid changes of Core records by an operational customizer. The so-called "Template protection". It contains an Include, which must be added to the generated maintenance view. Besides this, two events have been identified for activating this check to the maintenance view and the form included in the Include program has to be added to this event. The entry check is based on an authority check. For that reason the responsible people have to be authorized to maintain these tables.

The solution includes the following objects:

Source system developed in: DEC/100

PROG: /EON/CA_GEN_CHECK_AUTHORITY (includes the form for the event in the maintenance view)

SUSO: ZEON_CTABL (see also page 25 of this development guideline)

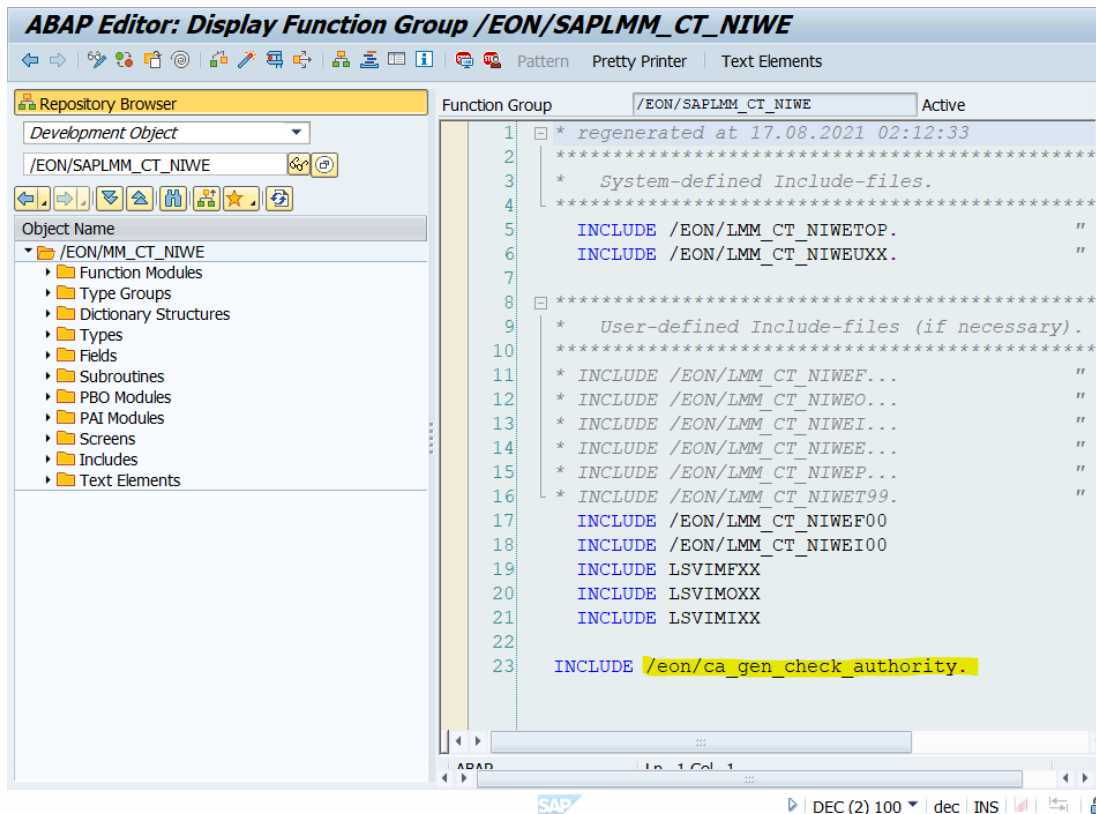
SUSC: ZEON (no Z, name to be reworked for S4U)

MSAG: /EON/CA_AUTH

DEVC: ZEON_CA (no Z, name to be reworked for S4U)

The include object has to be added to the main program of the function pool.

For example:



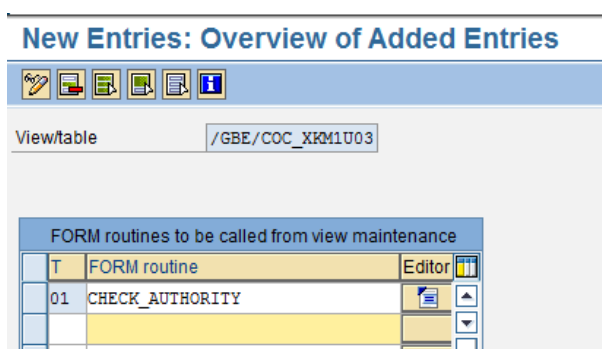
View Place of the additional authority check and coding below (outdated name)

```

*-----*
* Authority check to restrict changing template contents in the MU
* systems
* INCLUDE /EON/ca_gen_check_authority.
*-----*

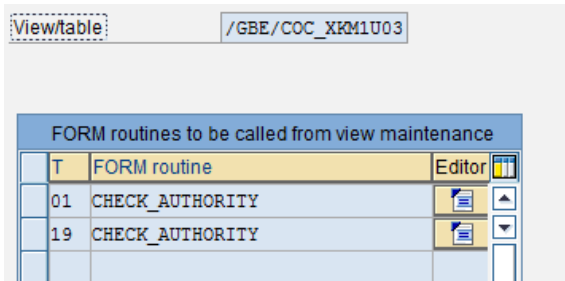
```

1. The event "01" has to be added to the maintenance view of the corresponding table

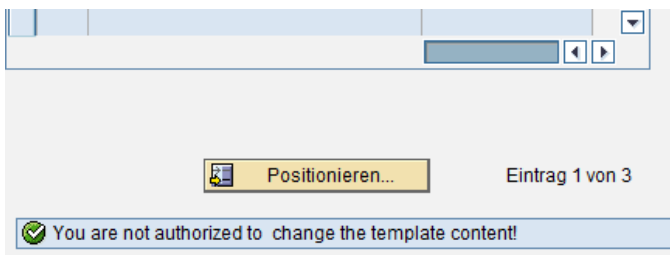


View View enhancement event 01 (outdated name)

2. The event "19" is processed during initialization: if the user does not have the authority to change the template content, he/she will see a status message to inform him/her:



View View enhancement event 19 (outdated name)



View Consequence of the check

The developer has to use always these objects by creating maintenance views in the corporate core area.

2.20. Screen Exit

SAP user-exit function module to transfer data to the customer implementation FROM SAP, e.g.
EXIT_SAPMIW00_008



PBO of the screen defined by SAP e.g. SAPLXQQM - 0101
(contains only the command for including a subscreen)



PBO of the new customer subscreen e.g. SAPLXQQM - 9100

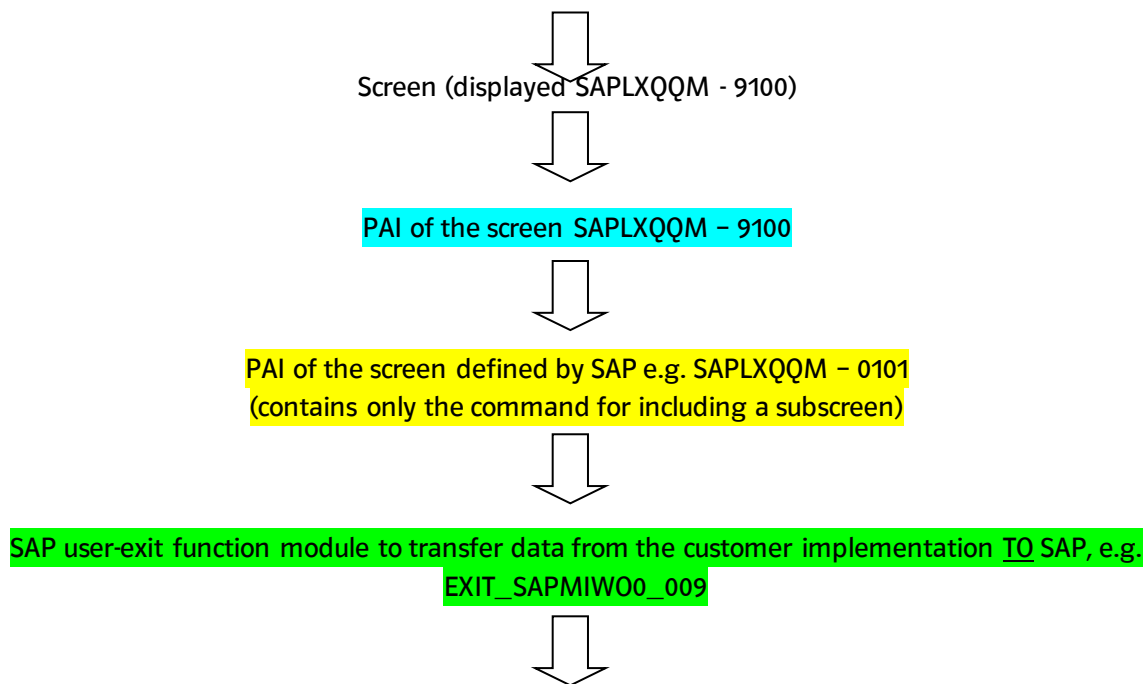


Figure: Screen Exit object flow

The customer own fields are to be defined on a 9+++ subscreen. Also screen exits have to be implemented using a steering table.

The PM solution (subscreen multi usage with different contents) from Innogy is accepted for grid screen exits to compensate the SAP limitations (only one customer subscreen allowed in transactions, but different subscreens required) as well as the E.ON solution from old GLOBE systems.

All developments are requested to get implemented in the respective new E.ON namespaces. The naming conventions and rules from that guideline applies for them.

2.20.1. Part screen process logic

The developer must:

- check whether the target screen contains the reserved place for the field to be added. If the place exists, he can implement his fields without a modification on this subscreen. (SE51 screen layout)
- extend the screen logic PBO (Process Before Output; for steering the display attributes and filling the existing field values from the database)
- extend the screen logic PAI (Process After Input; for checking the value against a check table or storing the possible changes in the database table(s))

2.20.2. Program Exit logic

To process the changes, the extended screen logic (extended by the customer application developer) leads the user to the programmed modules and functions or methods. These objects have to be created following the E.ON IT development guidelines and are already explained in the above description of the Program Exit itself.

2.21. Menu Exit

n/a.

2.22. Field Exit

A Field Exit is often used to implement additional check logic for a screen field. Field Exits can be implemented for each field, which is related to an ABAP data dictionary object (data element behind the screen field). The additional program logic is often covered in a Function Module belonging to the customer namespaces and processed (PAI) in the process logic of the screen. Most of the Field Exits found in S/4 systems are taken over from the legacy E.ON systems and belong to the non-scope area.

Following the SAP guidance field exits are obsolete in releases higher than 4.6C. Due to this fact no guidance for the implementation will be defined and field exits will not be created new anymore. In S/4 systems they can exist only based on a one to one transfer from former systems.

2.23. Customer screen control framework

We defined the rules for Core customer screen enhancements in an /EON/-implementation project with identical Interface definitions.

Core development (DEC)

Harmonized extension of SAP standard exits with customer fields

Simplified extension of SAP standard exits with customer fields

Tools for dynamic screen layout

Extensibility for other namespaces

For more details and consideration see the attached PPTX slides "S4U core development CSFW"



S4U core
development CSFW.

2.24. CMOD-Documentation

The developer has to create a project documentation in each customer CMOD project, which needs to provide at least these information:

- Business reason for implementation
- Short description of the functionality
- Components used in the solution
- The individual components must be documented according to their type.

2.25. Enhancements

The SAP Enhancement Framework provides a technology for enhancing source code units without modifying them.

Enhancements result in far less work during an upgrade compared to modifications. Enhancements are transport objects that can be stored in packages of their own unlike modifications, which are part of the object they modify. Since during an upgrade all SAP objects are replaced by the new version of these objects, after the upgrade all modifications are gone.

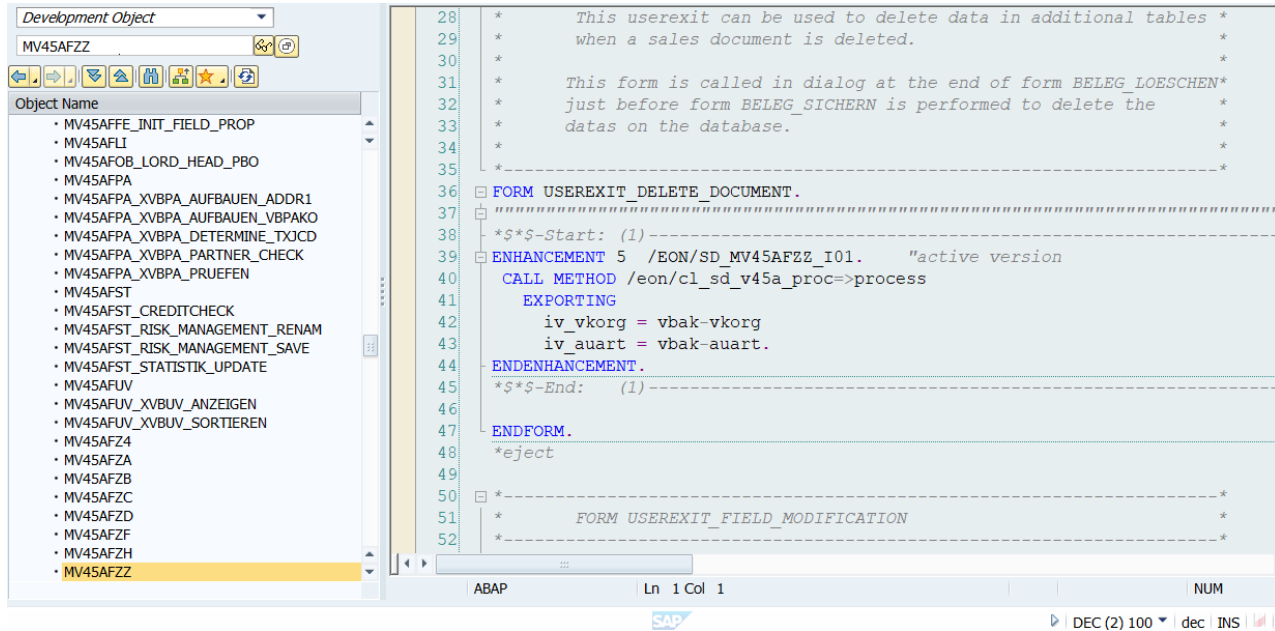
Manipulations/extensions to central posting modules (e.g. CO_ZV_ORDER_POST) are prohibited.

In contrast, customer enhancements as transport objects are never overwritten during an upgrade because they are stored in customer packages in the customer namespace. It may happen that the new version of a development object is no longer compatible with an enhancement, if for example an enhanced object is deleted or a data type used in the enhancement does no longer exist.

Every type of the enhancement framework is concerned (ENHO, ENHC, ENHS, etc.). Also implicit enhancements have to be extended with the new logic. That means every enhancement has to be controlled also by a control table! Due to the variety of the enhancements, a template logic can't be described in this document. The control of the enhancement has to be done by taking into account the object, the frame, the environment, the enhancement type.

Example 1:

Include MV45AFZZ contains dynamically called form routines (SAP standard) for the SD module. To control the different form routines via control table and to avoid the modification of this object, enhancements and the class logic have to be applied.



Some old enhancement options (see examples below), which were implemented in most of the source systems, might not be working in cloud based systems (like our S4U landscape) any longer and need to be replaced by another solution where appropriate. These replacement requirements are to be identified and implemented during transfers in the DET System.

New enhancement options for SAP S/4HANA

- SAP provides a new set of enhancement options (based on the proven BAdI technology) for the on-premise environment with SAP S/4HANA and for SAP S/4HANA Cloud
- For SAP S/4HANA Cloud, only the new BAdIs can be used with the Web-based ABAP editor and with the restricted ABAP for key users

Business Object	Old enhancement option	New enhancement option
Sales order	User exit MV45AFZZ	BAdIs in enhancement spot ES_SD_SLS_EXTEND (for example, sales header modification or sales document check before a save)
Purchase order	BAdI ME_PROCESS_PO_CUST	BAdIs in enhancement spot ES_MMPUR_PROCESS_PO_CLOUD (for example, purchase order header modification or final purchase order check before a save)

2.26. Authorization/Permission

2.26.1. Authorization/Permission group

The creation of a table maintenance dialogue requires the definition of an authorization group. The definition of the group name has to be aligned with the responsible person of the respective customizing function. The usage of &NC& (no authorization group) or even an empty field is not permitted.

The table maintenance has to be controlled by a maintenance transaction, ideally without using the SAP-Transaction SM30.

2.26.2. Authorization/Permission object

As soon as authorization objects are created in the customer namespace, the specifications and guidelines of SAP must be followed.

In order to standardize authorization maintenance and simplify subsequent role construction, the authorization objects as well as the authorization attributes have to be standardized and the functional specifications of the customizing managers have to be adhered to.

The following fields should always be present in the authorization object:

- protection of the program starting object, e.g. TCD (transaction code), SRV_NAME for FIORI tiles, etc.
- ACTVT (activity e. g. 01-create, 02-change, 03-display)
- Org structure fields, like: MM - EKORG, or SD – VKORG, buyers groups ... or other important and appropriate structure elements.

This allows generic authorization objects to be built up, which can be used by the developer colleagues in many ways. Thus, non-display transactions can easily be included in Display Roles, as ACTVT limits.

The name of the customer authorization object is to be classified correctly:

Core/Template = namespace or Yxxxx_FREETEXT, Non-Scope, Operations or Grid = namespace or Zxxx_FREETEXT.

xxx := SAP-Module

Group the Authorization objects according to SAP-Standard.

The developer must not forget to document the authorization object, it shall explain the meaning of it, the usages, etc..

In the case, that the renaming of existing authorization objects would lead to an unjustifiable effort, it is also possible to retain the name, simply change the original system and assign the object to a new package from one of the E.ON namespaces.

2.26.3. Prepared Developer Roles

The S4U IAM team has prepared the following roles for the developer in the systems of the two main level.

For the pure development activities in DET, DEC, DEO was the role X:PRO:DEVELOPER defined.

If in addition also customizing tasks are given to the developer in the D-systems, he needs to have the role X:PRO:CUSTOMIZER as well.

For testing purposes on the CEx system level is the project role X:PRO:DEBUG_REPLACE implemented, which allows also the debugging of developments during execution. Not allowed in the Q- and P-systems will be the usage of generic transactions, such as the SE16, SE38, SA38...

2.27. Other new development tools and areas

SAP provides periodically new development tools. Sometimes they use old transactions (SE11, SE51, ...), but combines them with new functionalities, simplifications, automatic steps processed in the background and so on.

2.27.1. EEWB (Easy Enhancement Workbench)

Is not expected to be used, as it focuses on CRM specific business objects. Furthermore it requires a similar high technical specification as the regular developments. The Wizzard used for implementation is always different, because its steps and options depend on the single business object, which shall be enhanced.

2.27.2. AET (Application Enhancement Tool)

The AET tool is a new tool for table and field extensions. This tool shall require less technical know-how as it provides a high automation and generation level during the definition. But it also requires an extended developer role set. The higher automation level with re-generation of unknown DDIC objects causes as well a higher risk to the system. The assignment of additional authorizations to a developer is to be confirmed by the system owner.

2.27.3. AIF (Application Interface Framework)

The AIF toolset uses own namespaces, which are independent from our development namespaces. They will be defined in the Customizing Guideline.

All development objects created for or used in an AIF interface, e.g. for data selection (outbound), mapping, data storage (inbound) or other purposes have to be defined according to the rules defined in this Development Guideline.

As soon as AIF is being used, has the requester to define an Archiving concept (for created IDOC's or XML-messages) to limit the database and table storage consumption to a minimum.

2.27.4. BRF+ (Business Rule Framework plus)

The BRF+ toolset *stands for Business Rule Framework Plus, which provides a comprehensive application programming interface and user interface for defining business rules. It enables you to define business rules without the need of writing ABAP code. The generated Business Rules can be incorporated into other SAP programs or Substitution/Validation rules 1).* All development objects created for or used in an E.ON own BRF+ application have to be defined according to the rules defined in this Development Guideline.

1) source: [Introduction to SAP BRF+/BRFplus & why is it so popular in SAP World !! | SAP Blogs](#)

2.27.5. CDS Views (Core Data Services)

A CDS View allows the customer to extract or fill in data directly from/to the database. That means they are not located on the application level and work around the role concept, which is as well valid only on application level.

Some Basic rules and priorities should be considered for the development and implementation of CDS views.

1. Continuity: ensure that the code runs correctly
2. Improve Performance: make code runs faster
3. Simplify: reduce complexity of data models and code

Other than that, we follow the rule set of SAP for the CDS-View creation and enhancement.

ABAP CDS - Language Elements

The following sections summarize the language elements of the [DDL](#) and [DCL](#) of the [ABAP CDS](#), arranged by topics.

- [DDL](#)
 - [CDS Views](#)
 - [CDS View Enhancements](#)
- [DCL](#)
 - [CDS Roles](#)
 - [CDS Access Policies](#)

In addition to the language elements shown here, can CDS source code also contain [annotations](#) for metadata.

DDL

CDS Views

Naming conventions

The following table contains the prefixes of the view types.

View type	Prefix
Basic interface view	I_

Composite interface view	I_
Consumption view	C_
Private View	P_
View enhancement	X_
Extension include view	E_
Abstract entity	D_

To avoid clashes with the defined naming convention for enhancement implementations, is it allowed to name the view as a combination of E.ON Namespace (5 digits) plus prefix (2 digits) plus free text. E.g. /ETN/I_

Enhancement- implementation#	<Namespace><Module>_<Characters>_ <2Digits>#
---------------------------------	--

Keywords and Additions

Language Element	Meaning
DEFINE VIEW ... AS	Defines a CDS view
WITH PARAMETERS	Defines input parameters of a CDS view
SELECT [DISTINCT] ... FROM	SELECT statement of a CDS view
INNER JOIN ... ON	Inner join in a SELECT statement
LEFT RIGHT OUTER JOIN ... ON	Outer join in a SELECT statement
ASSOCIATION ... TO ... AS ... ON	Defines an association for a path expression in a SELECT statement
1:	Attribute of a path expression
INNER ... WHERE	Attribute of a path expression
LEFT OUTER ... WHERE	Attribute of a path expression
KEY ... AS	Element of a SELECT list
\$EXTENSION.*	Elements of an enhancement in a SELECT list
WHERE ...	WHERE clause of a SELECT statement
GROUP BY ...	GROUP-BY clause of a SELECT statement
HAVING ...	HAVING clause of a SELECT statement
UNION ALL ...	UNION clause of a SELECT statement

Expressions

Operators	Meaning
MAX, MIN, AVG, SUM, COUNT	Aggregate expressions in a SELECT statement

<u>+</u> , <u>-</u> , <u>*</u> , <u>/</u>	Arithmetic expressions in a SELECT statement
<u>CASE</u> , <u>WHEN</u> , <u>THEN</u> , <u>ELSE</u> , <u>END</u>	Case distinctions in a SELECT statement
<u>CAST</u>	Casting expression in a SELECT statement
<u>=</u> , <u><></u> , <u><</u> , <u>></u> , <u><=</u> , <u>>=</u> , <u>BETWEEN</u> , <u>LIKE</u> , <u>IS NULL</u> , <u>NOT</u> , <u>AND</u> , <u>OR</u>	Relational expressions in a SELECT statement

Predefined Functions

Function	Meaning
<u>ABS</u> , <u>CEIL</u> , <u>DIV</u> , <u>DIVISION</u> , <u>FLOOR</u> , <u>MOD</u> , <u>ROUND</u>	Numeric functions in a SELECT statement
<u>CONCAT</u> , <u>LPAD</u> , <u>REPLACE</u> , <u>SUBSTRING</u>	String functions in a SELECT statement
<u>COALESCE</u>	Coalesce function in a SELECT statement
<u>CURRENCY CONVERSION</u> , <u>UNIT CONVERSION</u> , <u>DECIMAL SHIFT</u>	Conversion functions in a SELECT statement

CDS View Enhancements

Keywords and Additions

Language Element	Meaning
<u>EXTEND VIEW ... WITH</u>	Enhancement of a CDS view with an CDS view enhancement

DCL

CDS Roles

Keywords and Additions

Language Element	Meaning
<u>DEFINE ROLE ... GRANT SELECT ON</u>	Defines a CDS role
<u>WHERE</u> , <u>AND</u> , <u>OR</u>	WHERE clause in the definition of a CDS role
<u>... = ASPECT pfcg auth ...</u>	Condition for mapping roles
<u>... = <> < > <= >= LIKE IS NULL ASPECT ...</u>	Condition for assignment roles
<u>... = <> < > <= >= LIKE IS NULL ...</u>	Literal condition

CDS Access Policies

Keywords and Additions

Language Element	Meaning
<u>DEFINE ACCESSPOLICY</u>	Defines a <u>CDS access policy</u>
<u>DEFINE ASPECT ... AS</u>	Defines a <u>CDS aspect</u>
<u>SELECT FROM</u>	<u>SELECT statement of an CDS aspect</u>
<u>WHERE \$user IN</u>	<u>WHERE clause of the SELECT statement of an CDS aspect</u>

Source:

https://help.sap.com/doc/abapdocu_740_index_htm/7.40/en-US/index.htm?file=abencds_language_elements.htm

For CDS Views is the definition of a Data Control Language, DCL based authority check mandatory. The ABAP CDS View Access Control with DCL provides the developer the way to check the authorization of a user and limits the access based on PFCG authority objects.

In case the developer defines a joined CDS view for more than one table, she/he has to consider the org. structure level element related authorization checks (e.g. FI = BUKRS, MM = EKORG, WERKS, CO = KOKRS) for all single tables. If no standard authority check object exists for a table, is the developer responsible to create one in SU21 (namespace / naming convention depends on development system and are to be used). This is important to guarantee, that no cross organization data and no GDPR relevant data can be extracted without having the respective authorization.

Examples for authorization objects for SAP Standard tables are:

Display Authorization Object	Display Authorization Object	Display Authorization Object																				
Object F_BKPF_BUK	Object M_BEST_EKO	Object K_CSKS																				
Text Accounting Document: Authorization for Company Codes	Text Purchasing Organization in Purchase Order	Text CO-CCA: Cost Center Master																				
Class FI Financial Accounting	Class MM_E Materials Management: Purchasing	Class CO Controlling																				
Author	Author	Author																				
Authorization fields	Authorization fields	Authorization fields																				
<table><tr><th>Authorization Field</th><th>Short Description...</th></tr><tr><td>BUKRS</td><td>Company Code</td></tr><tr><td>ACTVT</td><td>Activity</td></tr></table>	Authorization Field	Short Description...	BUKRS	Company Code	ACTVT	Activity	<table><tr><th>Authorization Field</th><th>Short Description...</th></tr><tr><td>ACTVT</td><td>Activity</td></tr><tr><td>EKORG</td><td>Purchasing organization</td></tr></table>	Authorization Field	Short Description...	ACTVT	Activity	EKORG	Purchasing organization	<table><tr><th>Authorization Field</th><th>Short Description...</th></tr><tr><td>KOKRS</td><td>Controlling Area</td></tr><tr><td>KOSTL</td><td>Cost Center</td></tr><tr><td>ACTVT</td><td>Activity</td></tr></table>	Authorization Field	Short Description...	KOKRS	Controlling Area	KOSTL	Cost Center	ACTVT	Activity
Authorization Field	Short Description...																					
BUKRS	Company Code																					
ACTVT	Activity																					
Authorization Field	Short Description...																					
ACTVT	Activity																					
EKORG	Purchasing organization																					
Authorization Field	Short Description...																					
KOKRS	Controlling Area																					
KOSTL	Cost Center																					
ACTVT	Activity																					

The definition of the appropriate authorization object is with the functional team and to be described in the functional and technical specification document.

An example for a CDS View with Access Controls is the following implementation. The Core Data Service starts with the definition of the access control area.

- ✓  Core Data Services (6)
 - ✓  Access Controls (1)
 - >  /EON/ARBPO_OV Ariba PO Overview - Acces Control
 - ✓  Data Definitions (5)
 - >  /EON/ARBPO_ADPO Ariba PO Overview - ADPO
 - >  /EON/ARBPO_ALL Ariba Po Overview - All
 - >  /EON/ARBPO_CUST Ariba PO Overview - CUST
 - >  /EON/ARBPO_EKPO Ariba PO Overview - EKPO
 - >  /EON/ARBPO_ODAT Ariba PO Overview - ODATA Service
- ✓  Dictionary (14)

The access control itself is defined as follows:

```
[DEC] /EON/ARBPO_OV ⓘ
1 @EndUserText.label: 'Ariba PO Overview - Acces Control'
2 @MappingRole: true
3 define role /eon/arbpo_ov {
4     grant
5         select
6             on
7                 /EON/ARBPO_ODAT
8                 where
9                     ( bsart ) =
10                     aspect pfcg_auth ( M_BEST_BSA,
11                     BSART,
12                     ACTVT = '03'
13                     )
14     and ( ekorg ) =
15     aspect pfcg_auth ( M_BEST_EKO,
16     EKORG,
17     ACTVT = '03'
18     )
19     and ( werks ) =
20     aspect pfcg_auth ( M_BEST_WRK,
21     WERKS,
22     ACTVT = '03'
23     );
24 }
```

Other naming conventions and Namespaces for CDS Views need to fulfill the rules defined in this Development Guideline.

2.27.6. ABAP in SAP Cloud Platform (SCP or BTP)

ABAP developments in the SAP Cloud Platform provides additional opportunities to develop customer functionalities for SAP S/4 HANA cloud systems. These customer own functionalities access the backend SAP systems via remote API's. The usage of remote API's allows only access to whitelisted ABAP platform objects and restricts the available technology components, for example no SAP GUI access and a restricted ABAP Language scope. The restricted ABAP language scope excludes classical development transactions, like SE80. It excludes statements, which are not compliant with secure cloud operations, such as direct access to the file system. In addition, ABAP statement variants which have already been deprecated have been removed from the language scope. Modification concepts of vendor software, familiar from SAP on-premise solutions, like Customer-Exits or Enhancements are not feasible in the Cloud. Beside these SAP controlled limits and security rules underlie Cloud Platform developments the same rules as on-premise ABAP developments, described in this document.

The EAM Team and the Team from **SAP CCE – SAP Competence Center of Excellence** are about to create an own development guideline for developments in this environment.

2.27.7. JAVASCRIPT recommendations

In regards to javascript developments we follow the approach defined by the LEW (Lechwerke) in the first step. This might be changed /enhanced as soon as an E.ON wide javascript development guideline will be defined / in place.

The „Golden Rules“ collect the most important Guidelines:

1.	Methods and functions shall be as small as possible and fulfill only one single task. They should have only 2 to 3 Input-Parameter. The usage of complex Config-Objects (mParam) as Input-Parameter have to be avoided.
2.	Reduce dependencies to the base project to a minimum. If possible, re-implementations and updates should be made directly in the respective project.
3.	Design objects are meaningful and descriptive. Use a uniform naming for the same objects (e.g. cost estimate).
4.	Comments should be well worded and concise. You want to describe WHY something was implemented in this way and not WHAT the code does.
5.	For requests to the backend service: As little possible as possible, as much as necessary. Check carefully whether the request is really necessary.
6.	For consistency of the application and consistency between the different applications, major changes need to be made with the lead developers.
7.	Service entities and their properties should be bound to the UI controls using data binding via the OData model.
8.	For applications with complex application logic (e.g. deep insert) a separation after "Separation of Concerns (SOC)" should be carried out. ¹
9.	The use of promises (async, await) in connection with try catch blocks is preferable over callback functions.
10.	A "throw error" statement must always be handled in a catch block.

Principles

UI5/Javascript developments in the S4U and EAM context should follow the following principles.

Principle	Explanation and priority
SOLID	• Single Responsibility • Open Closed • Dependency Inversion
KISS	Keep it simple and stupid

¹ SOC: In the context of a UI5 application, the MVC pattern is the structuring pattern to differentiate between model, view, and a controller. However, the MVC does not describe a strict separation of modules within a controller. For complex applications, a separation between business logic, flow logic and network handling is necessary. The SOC refers to the separation of responsibilities within the controller. For example, business logic should be handled in an entity module and network handling in a request module. According to Law of Dementer, the controller should only talk to the entity. This increases decoupling and reduces dependence. In addition, the complexity of the controller is reduced.

DRY	Don't repeat yourself
Law of Dementer	According to Information Hiding, objects should only communicate with directly adjacent objects.
GoF Pattern	Design patterns for the creation, behavior, and structuring of objects

Naming conventions:

- Meaningful and purpose-descriptive names (preferably too long than too short)
- Only English names and descriptions
- Uniform names for same objects
- Use of **Camel** Case notation (e.g. oltemList)
- Use of prefixes (Hungarian notation):
-

Sample	Type
s Id	string
o DomRef	object
\$ DomRef	jQuery object
i Count	int
m Parameters	map / assoc. array
a Entries	array
d Today	date
f Decimal	float
b Enabled	boolean
r Pattern	RegExp
fn Function	function
v Variant	variant types
p Dialog	promise

Best Practices

- Please refer also to the
 - SAP Javascript best practices under:
<https://sapui5.hana.ondemand.com/#!/topic/eded636b85584cd586b1fe231d2b5dac>
 - SAP best practices under:
<https://sapui5.hana.ondemand.com/#!/topic/28fcd55b04654977b63dacbee0552712>
- No use of global variables at the component level
- Reduction of member variables within the module/controller
- Methods and functions should be as small as possible and do only one task. You should have only 2 to 3 input parameters and do not use complex Config objects (mParam) as input parameters.
- For applications with complex application logic (e.g. tree flap assistant, deep insert) a separation according to SOC should be carried out
 - To use a Request/Entity Module:
 - Request Module: Aggregation of all HTTP Requests and Responses
 - Business Object / Entity Module: Dependency to Request Module. Represents the function of a business object
 - Demarcation: Simple application with simple/few backend calls should not use this model

- Layout preferably in XML fragments and views (no UI creation in the controller)
 - Keep controllers as small as possible and outsource other logic to modules (avoidance of god classes)
- Orientation to the SOLID principles
 - E.g. Decomposition/Modularization of classes / logic
 - Avoid static classes (exception: Factories)
 - Reduce Utilities
- Reduction of dependencies
 - Reduce dependencies to the base project to a minimum. If possible, re-implementations and updates should be made directly in the respective project.
- Define and use constants objects (no hard coded Strings)
- In the following cases, a generic in-house development is preferable to the SAPUI5 Programming Model (MVC):
 - If the component is expected to be subject to frequent extensions in the future
 - If the behavior of the components depends on the backend (e.g. editing process)
 - When double coding can be reduced/avoided
 - The choice between a generic/complex solution versus a standard UI5-MVC solution is supported by lead developers in the design process.
- Do not use private member objects/variable/functions of other objects
- When leaving a view, it must be ensured that it is cleaned up correctly and re-initialized (e.g. events de-register, destroy custom objects, ...).
- The following lifecycle hooks should be used for this purpose:
 - **onBeforeHide Event on the NavContainer:**
Called as soon as a navigation has been initiated and the child controls of the page are hidden

```
onInit: function () {  
    ...  
    this.getView().addEventDelegate({  
        onBeforeHide: (oEvent) => {  
            //When leaving the page, it is re-initialized  
        }  
    });  
},
```

- **onExit Event am Controller:**

Called when the page is destroyed. For example, when a back navigation is triggered via the FLP Home button.

```
onExit: function () {  
    this._unsubscribeBmAssistEvents();  
    this.getOwnerComponent().getModel('BaumappeModule').unregisterEvents();  
},
```

comments

- Comments should not be a supplement to bad coding. Better than comments are a clear naming and clean coding.

- Comments should be well worded and concise. You want to describe WHY something was implemented in this way and not WHAT the code does.
- If comments are necessary, then JSDoc commands should be used.
- No commented-out code. Traceability of changes via Git possible.
- Comments have to be written exclusively in English.

organizational

- For consistency and consistency between the different applications, major changes must be agreed with the lead developers.

Java Script

- Use EcmaScript 6 features as much as possible
 - Compatibility with different JS-runtimes and browsers guarantees a transcompiling process in CI/CD processes
 - let instead of var
 - Use of promises (async, await) in connection with try catch blocks. The use of callback functions must be dispensed with if possible (Callback-Hell)
- sap/base/log instead of console.log
- Comparison operators
 - Use "===" instead of "=="
 - Use "!== " instead of "!="
- Class/Module
 - Member variables should be declared privately in the constructor
 - Example: "this._bReady = false;"
 - Instance methods to be defined using the Prototype property
 - Example: "MyClass.prototype.myMethod = ()=>{"
 - Static methods should be defined directly at the class
 - Example: "MyClass.myStaticMethod = ()=>{"
 - Private member variables and methods are defined by the name prefix "_"
 - Example: "this._bFinalized"
 - The use of setTimeout is to be dispensed with, depending on the environment, different results may occur. Better use callback functions or promise objects to process asynchronous

Exception-Handling

- Always handle "throw error" instructions in the catch block.
- Message Manager must exist in every app
- View unique error messages for the user (Message Manager)

Networking

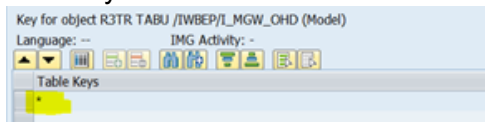
- Intelligent use of \$batch (e.B. lazyloading)
- For requests to the backend service: As little possible as possible, as much as necessary. Check if Request is really necessary.
- Use the \$select option in the Odata request to reduce the number of attributes

SAPUI5/Data-Binding

- UI texts are off-exploited to an i18n file.
- Service entities and their properties should be bound to the UI controls using data binding via the OData model/JSON model.
 - An explicit use of ODataModel.read should only be used in relevant cases due to compatibility reasons from OData V2 to OData V4.
 - Explicit value assignment via the setter methods of the UI controls should be avoided (e.g. setValue)
- Use annotation-binding for labels, Maxlength, Tooltip, etc.

General:

- Unified Folder Structure App Cross-App
- Configuration records from the following control tables need to be inserted manually into transports.
 - /IWBEP/I_MGW_SRG
 - /IWBEP/I_MGW_SRH
 - /IWBEP/I_MGW_SRT
 - /IWBEP/I_MGW_OHD
 - /IWBEP/I_MGW_OHT
- The table keys for these tables **must not be “*”**, as this would overwrite and delete records in the common used systems CEC/QEC/PEC and also CEO/QEO/PEO.
- Not allowed anymore are records like this:



-
- The service name must be defined in the table key. Common because:
 - CEC/QEC/PEC get table records from DEC and DEA.
 - CEO/QEO/PEO get table records from DEC and DEO.

2.27.8. JAVA recommendations

Java developments are requested to consider and fulfill the Security checkpoints listed in the embedded EXCEL spreadsheet.



E.ON Application
Security Java - Secur

The latest version of the document is published in E.ON's Connect:

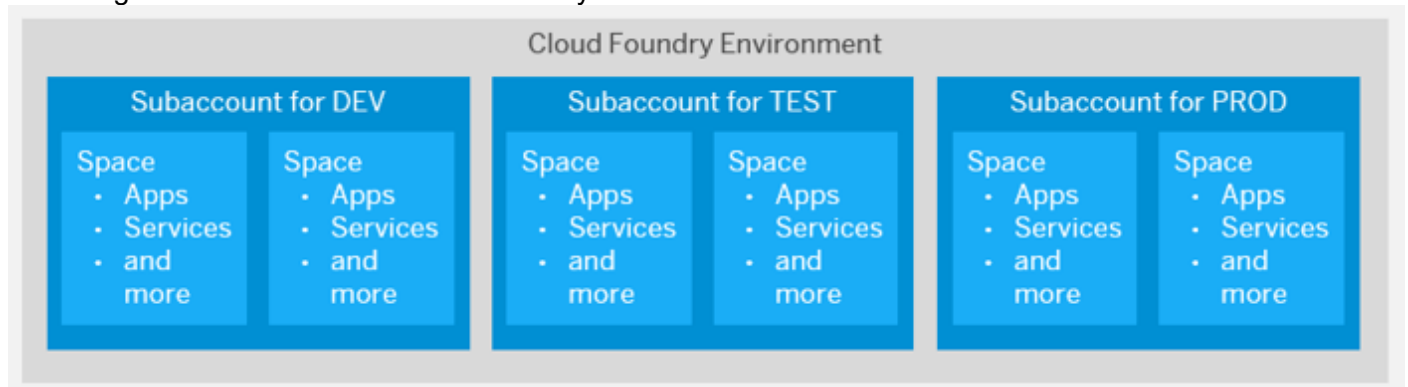
<https://connect.eon.com/workspaces/ws-application-security-e-on/apps/content/app-guidelines7e#SecureCoding>

JAVA developments need to consider the XSRF Protection Guidelines described in the SAP Notes 1544240 - Unauthorized use of application functions in SAP NW Portal. In the attached files to this note you can find:

1. EP_XSRF_Protection_Guide: A PDF presentation explaining how to use the XSRF protection framework provided by the Java engine in order to protect a portal application.
 2. com.sap.portal.security.csrf.example: A Sample portal application.
- and 1450166 - explaining how to use the XSRF Protection Framework provided by the Java engine.

JAVA Developments connected with S4U and EAM need to be implemented in the GLOBAL Account of Innogy in the Cloud Foundry in AZURE.

Following Subaccount structure is mandatory:



The following Subaccounts will be defined for usage in S4U:

- S4U-Dev
- S4U-Cons
- S4U-Qas
- S4U-Prod

The individual IT change manager is requested to organize required Directories and SPACES for the separation and collection of developments together with the Cloud Service platform team

(email to: VLSAPCloudPlatform@eon.com). A request form is available and requires these information from the requester:

Required information

Mandatory

- Cost centre
- Project name
- Subaccount name
- Plattform (Cloud Foundry, Kyma, ABAP, NEO,)
- Environment (Prod, Dev, Test..)
- Provider (AWS, Azure)
- Services / Entitlements
- Technical contact – Subaccount admin
- Roles and Role collections

Optional

- Architecture (draft)
- CloudConnector + Backend
- Users
 - Cloud Foundry – email address connected to the S-User
 - NEO – S-User
 - IAS – KID and email address
- Spaces



2.28. Internet Transaction Services (ITS)

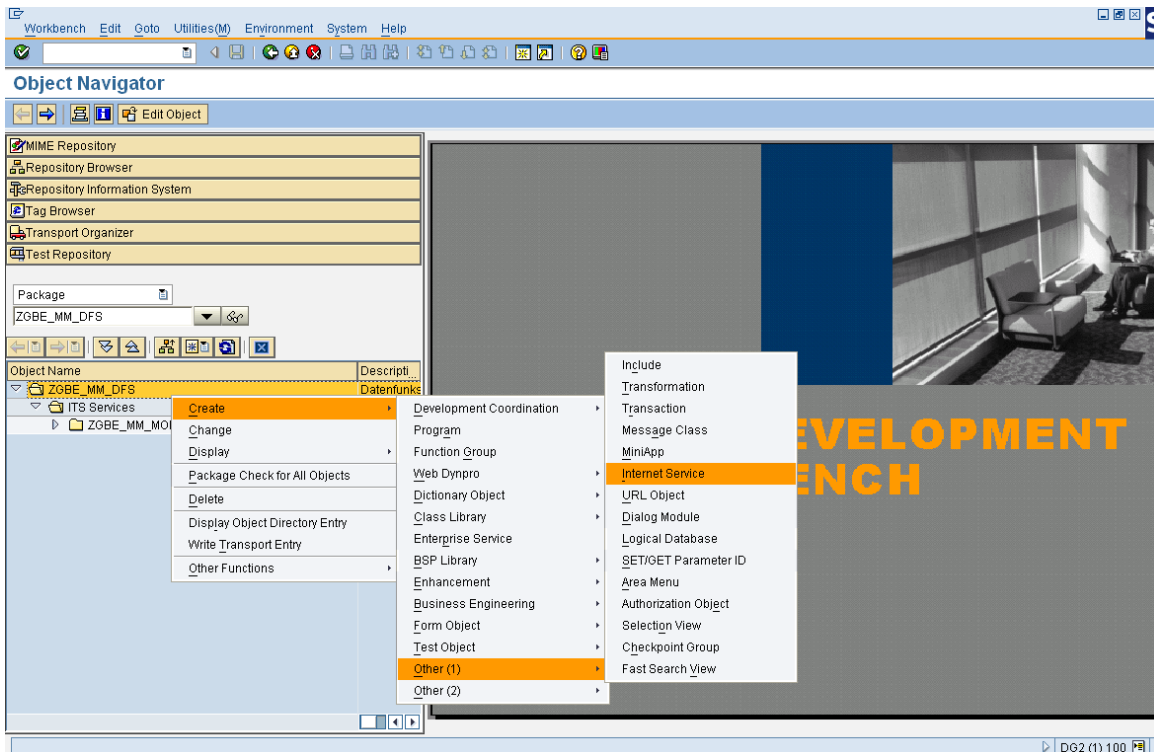
Even for the S/4 United core functionalities new internet transaction services (ITS) can be required. Limitations within the SAP standard as far as the use of customer namespaces - like /EON/ - are concerned, can require to use the customer namespace Z for this core application.

2.28.1. Prerequisites

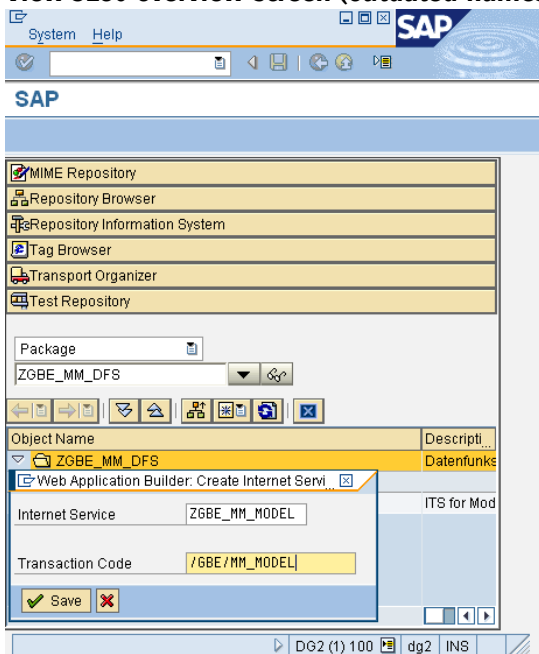
- A SAP Web AS ABAP application is necessary for the template generation. The screen layout of the application must be designed for mobile devices and their limited visualization functions. It has to be created with transaction SE80.
- The maintenance size of the screen must be adjusted to the screen size of the mobile device. The HTML template generator uses the maintenance size of the screens (number of rows / columns) to build an HTML table with a character matrix.
- The application should use only screen elements that are supported by ITSmobile. For a list of supported elements, see Supported Screen Elements.
- For the access to the SAP system a corresponding URL has to be defined.
URL definitions have to be requested from the Basis team. The URL must be defined like (outdated)
https://dg2.apps.eon.com/sap/bc/gui/sap/its/my_requirement/dfs00?sap-client=100
Because of the fact, that all GLOBE systems use the same WEB dispatcher, the URL has to be defined just once and can be used for each other system as well. Adoption: the system ID and maybe the client have to be changed.

2.28.2. Creation of the Internet Transaction Service with SE80

The ITS has to be assigned to an own new development package in the core- or operations- or network- or Z-namespaces. **This is to be reworked / rechecked for S/4 United systems:**
The service is created within the Z-namespace, even that it is part of the S/4 United core, because for the name of Services (Transaction SICF) SAP doesn't support customer own namespaces containing special signs (ex. / , \$ and so on).

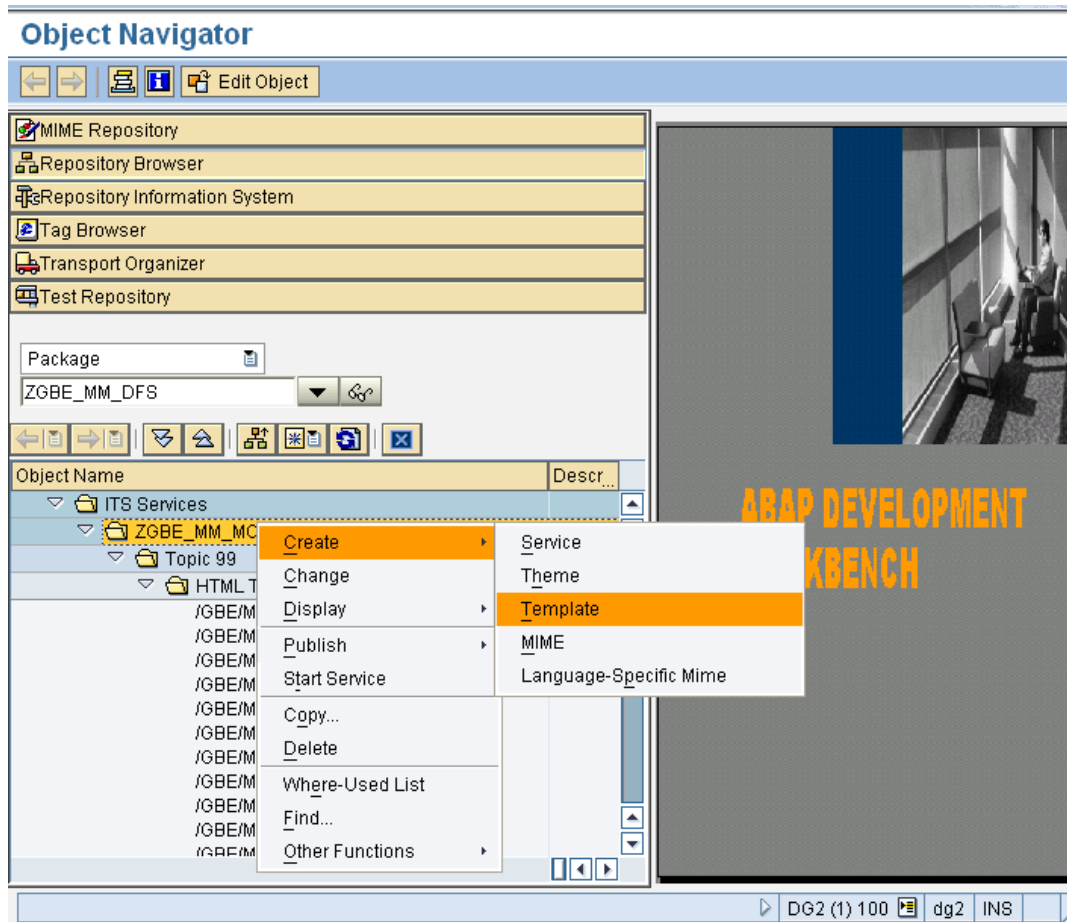


View SE80 overview screen (outdated names)



View assignment information (outdated names)

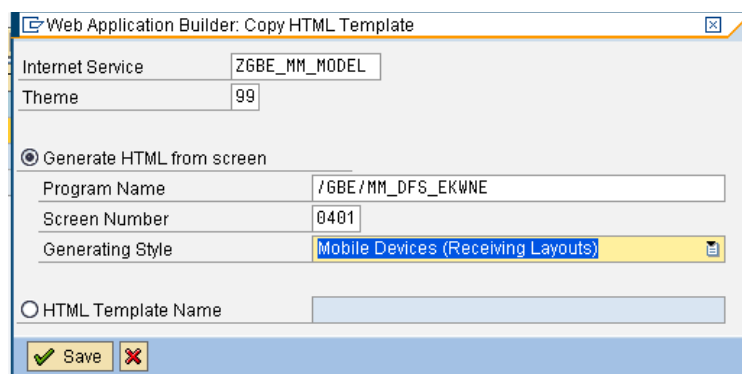
The ITS is a structured object with different themes.



View Template creation path (outdated names)

Maintain the field Theme (we defined 99 as in the template itsmobile01).
Maintain also the fields:

- program name (Name of used program for ITS),
- Screen Number (for every used screen number a template must create) and
- Generation Style (in our case Mobile Devices (Receiving Layouts)).



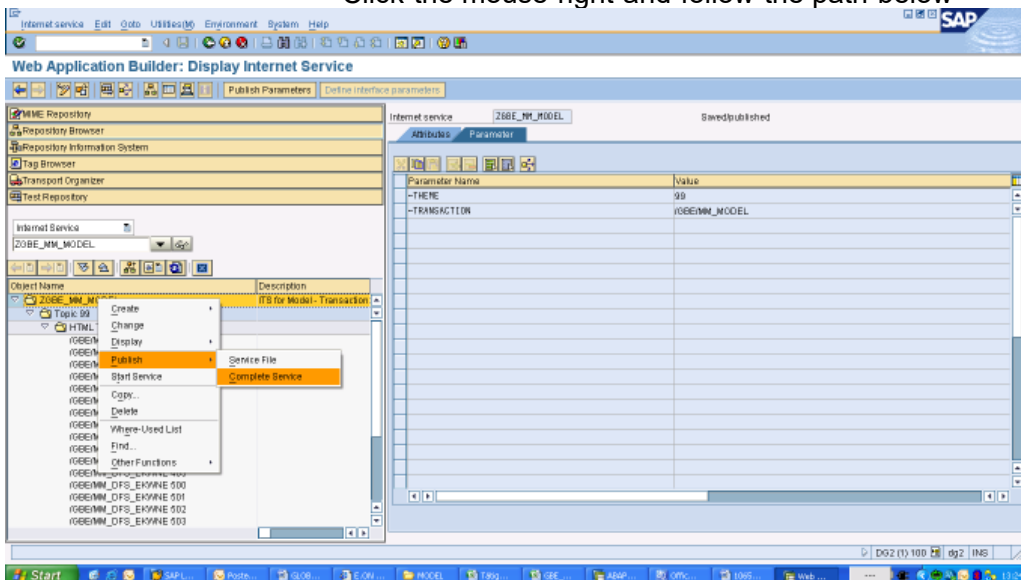
View detailed view (outdated names)

HTML's can be copied from other existing template, choose HTML Template Name. Define name and add the HTML coding manually (e.G. for the device include)

Before the service can be used, the service must be published.

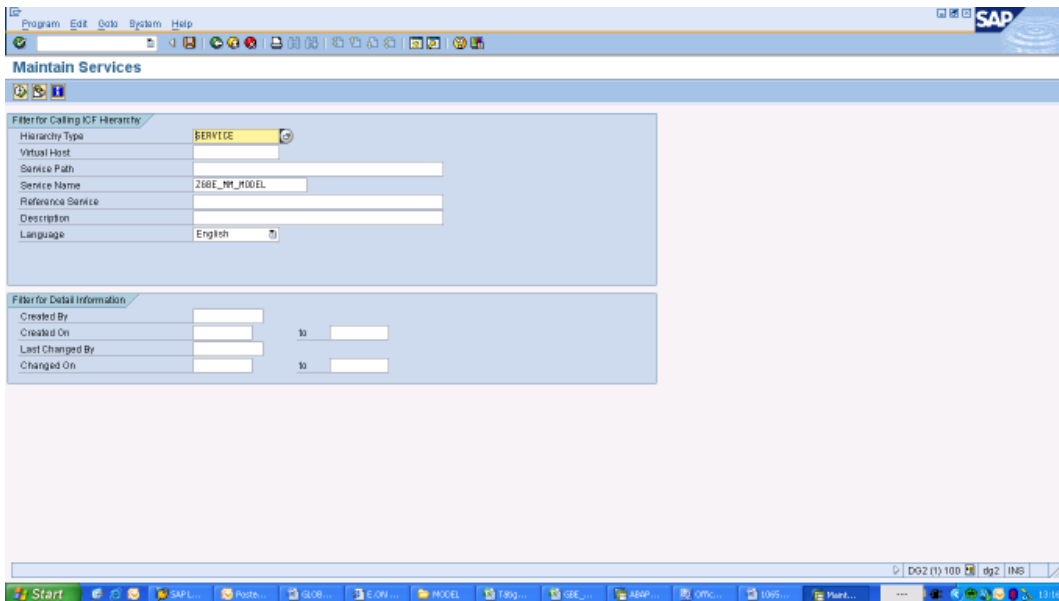
To do that, open the service in SE80.

- ➔ Double click on the service name
 - Click the mouse right and follow the path below



View path to publish (outdated names)

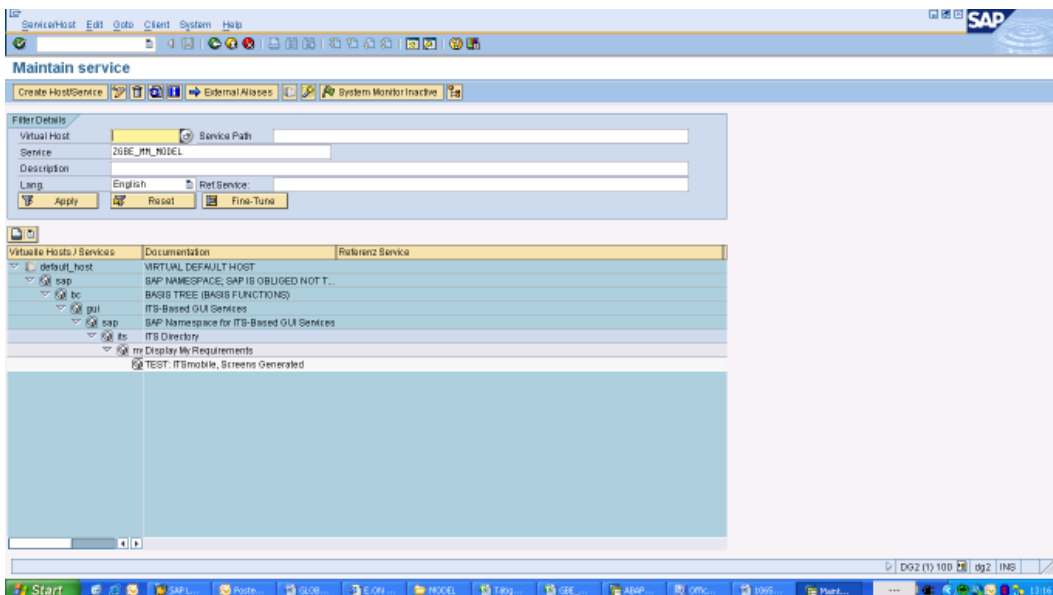
2.28.3. Definitions in the transaction SICF



View 47 SICF starting screen (outdated names)

The service ITSMOBILE01 was copied under the path:

default_host -> SAP -> bc -> gui -> SAP -> its -> my_requirement. After copy the Service ITSMOBILE01 be rename to **ZEON_MM_MODEL**. (Z-namespace a "Must"? To be checked for S4U)



View SICF service structure overview (outdated names))

When double clicking on the service **ZGBE_MM_MODEL** (Z-namespace a “Must”? To be checked for S4U), the following screen gets opened.

Service Edit System Help

Create/Change a Service

Path /default_host/sap/bc/gui/sap/its/my_requirement/

Service Name DF500 Service (Active)

Lang. English Other Languages

Description

Description 1 TEST: ITSmobile, Screens Generated and Modified

Description 2

Description 3

Service Data Logon Data Handler List Error Pages Administration

Service Options

☐ Web Service

☐ Do Not Include Inherited Settings

Load Balancing

SAP Authoriz.

Session Timeout 00:00:00 (HH:MM:SS)

Compression Not specified

Interactive Options

GUI Link Yes

Support Accessibility Not specified

GUI Configuration

DG2 (1)

View SICF service editor (outdated names)

Push the button: **GUI Configuration** to open the screen for the parameter maintenance.

Service Edit System Help

Create/Change a Service

Path /default_host/sap/bc/gui/sap/its/my_requirement/

Service Name ZGBE_MM_MODEL

Lang. English

Description

Description 1 TEST: ITSmobile, Screens Generated

Description 2

Description 3

Service Data Logon Data Handler List Error Pages

Service Options

☐ Web Service

☒ Do Not Include Inherited Settings

Load Balancing

SAP Authoriz.

Session Timeout 00:00:00 (HH:MM:SS)

Compression Not specified

Interactive Options

GUI Link Yes

Support Accessibility Not specified

Display Service Parameters

Parameter Name	Value
~THEME	99
~TRANSACTION	/GBE/MM_MODEL
#MOBILE	This is a mobile service
~ITSMOBILE	1
#SOURCES	Template search sequence
~SOURCES	ZGBE_MM_MODEL
#POPUPS	Show popups, default is no
~POPUPS	1
#ITSMOBILEDEVICEINCL	Include a device specific html
~ITSMOBILEDEVICEINCL	DEVINCLUDE_NAURTECHK81
#MESSAGE	SOUND Plays a background sound if a message appears
~ITSMOBILEMSG SOUND	1
#SYSTEM	MODAL POPUPS Shows express mails / system message
~SHOW_SYSTEM_MODAL	1

DG2 (1) 100 dg2 INS

View SICF service parameter editor (outdated names)

Very important is the parameter ~ITSMOBILEDEVICEINCLUDE. This parameter has to contain the name of an existing device template in the SE80 ITS. For MODEL the DEVINCLUDE_NAURTECHCK61.HTML is defined in the service ZGBE_MM_MODEL (Z-namespace a "Must"? To be checked for S4U) as a copy from the itsmobile01 template.

The parameter ~SOURCES defines which Internet transaction service (SE80) is read to get the steering information (screens, devices, etc.)

An overview of the parameter can be found behind the link:

http://help.sap.com/saphelp_dimp50/helpdata/EN/bd/f8cf3f06bd3946e10000000a114084/content.htm

2.28.4. Changes on scanner side

Changes on the scanner side have to be identified, planned and done by the business. Details have to be described in the functional design or in implementation guides from the former solution.